The elephant random walk with rare reversals: a Luria–Delbrück approximation and the shape of the law

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Abstract

In the elephant random walk each step copies a uniformly chosen earlier step and reverses it with probability ε. We let $\varepsilon \to 0$ as the number of steps N grows, with $N\varepsilon$ of order $\log N$. Let M_N be the number of steps that disagree with the first step. We prove an explicit total variation bound between the law of M_N and a truncated Luria–Delbrück law, the compound Poisson law with jumps d at rates $N\varepsilon/(d(d+1))$. A Landau local limit theorem with an explicit rate follows. We then derive the shape of the law of the position. Every mode lies at distance $x \log x + \lambda_{\mathrm{L}} x + O(\sqrt{x \log x})$ from an end, where $x = N\varepsilon$ and $\lambda_{\mathrm{L}} \approx -0.2228$ is the Landau mode. For every N and $\varepsilon < \frac{1}{2}$ the second difference of the law changes sign at most 4 times. For even N the probabilities of the origin and of each end cross exactly once, at $N\varepsilon = 2\log N - \log\log N - \log 16 + o(1)$. There, for every even $N \geq 8$, the origin is the unique interior minimum and the law has exactly 2 symmetric intervals of modes, by certified computations at finitely many lengths. The sign theorem passes to the superdiffusive limit law at fixed memory $a = 1 - 2\varepsilon \in (\frac{1}{2}, 1)$. With a fair first step this law has at most 2 modes for every a, and exactly 2 for $a \geq \frac{39}{40}$. By a computer-assisted proof it is unimodal for $a \leq 0.668812860014597$ and bimodal for $a \geq 0.668812860014602$. Up to flat modal intervals, this settles the fair-start case of a conjecture of Guérin, Laulin, Raschel and Simon.

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1 Introduction

In the elephant random walk of Schütz and Trimper [1], each step copies a uniformly chosen earlier step and reverses its sign with probability ε. The first step is fair. We study rare reversals. The reversal probability ε tends to 0 as the number of steps N grows, and $x = N\varepsilon$ is of order $\log N$. So reversals become rarer even though their expected number $x - \varepsilon$ grows. Most limit theorems instead fix ε [4, 5, 6, 7, 9].

Let M_N be the number of steps that disagree with the first step. Our main result is a total variation bound of order $x^2 \log^2 N/N$ between the law of M_N and a truncated Luria–Delbrück law, namely the compound Poisson law with jumps $d \leq N-1$ at rates $x/(d(d+1))$. From it we derive a Landau local limit theorem with an explicit rate (Theorem 1.1). The modes, the shape of the

law and the crossing below are consequences. Every mode lies at distance about $x \log x - 0.2228\, x$ from one of the two ends (Theorem 1.2). The second difference of the law changes sign at most 4 times (Theorem 1.4).

The count M_N is the resistant-cell count in Bokes, Hlubinová and Singh's reversible fluctuation assay [22], with one sensitive founder, uniform parent choice and equal switching probabilities at birth. Their first-switch and Pólya-urn calculation already gives the first-order center coefficient $4/(N+2)$ [22, Eqs. (6)–(10)]. Their matched asymptotics give the Landau shape for one sensitive founder [22, Eqs. (50)–(52)], and two Landau densities for a random start [22, Eqs. (58)–(60)]. These results are formal. Our contribution is quantitative. We give explicit error bounds, and from them exact crossing and shape results.

For even N we compare the probability of the origin with that of each end $\pm N$. Near $\varepsilon = 0$ the ends are more likely, while at $\varepsilon = \frac{1}{2}$ the steps are independent and the origin is more likely. We prove that the two probabilities cross exactly once, and we locate the crossing at $x = 2\log N - \log\log N - \log 16 + o(1)$ (Theorem 1.3). At the crossing, for every even $N \geq 8$, the origin is the unique interior minimum and the law has exactly 2 symmetric intervals of modes (Theorem 1.4). The persistent walk, which copies the last step rather than a uniform earlier step, crosses at about $\log N$ switches [11].

The sign theorem holds for every N and every $\varepsilon < \frac{1}{2}$, so it also applies at fixed memory $a = 1 - 2\varepsilon \in (\frac{1}{2}, 1)$, where $(2A_N - N)/N^a$ has a superdiffusive limit law. With a fair first step this law has at most 2 modes for every a, and exactly 2 for $a \geq \frac{39}{40}$ (Theorem 1.5). A computer-assisted proof shows that it is unimodal for $a \leq 0.668812860014597$ and bimodal for $a \geq 0.668812860014602$ (Theorem 4.4). Up to flat modal intervals, this settles the fair-start case of a conjecture of Guérin, Laulin, Raschel and Simon [7, Remark 2.1(b)], with the threshold located to within $3.3 \cdot 10^{-15}$. A weak limit connects our Landau law to this limit law as the memory tends to 1 (Corollary 4.1).

For comparison of conventions, the original memory parameter is $1 - \varepsilon$, the parameter a of [7] is $1 - 2\varepsilon$, and the broadcasting flip probability is ε. The broadcasting tree representation is recalled in Lemma 3.1 [2, 3].

1.1 Setting and notation

The walk takes N steps $X_1, \ldots, X_N \in \{+1, -1\}$, where $+1$ means right. Let $A_N = \#\{i \leq N : X_i = +1\}$ be the number of right steps, and write $P_N(k) = \mathbb{P}(A_N = k)$ for the probability of the value k. The position $2A_N - N$ is at the origin when $A_N = N/2$ and at an end when $A_N \in \{0, N\}$. Unless we say otherwise the first step is fair, $\mathbb{P}(X_1 = +1) = \frac{1}{2}$, and the rules treat the two directions alike, so $P_N(k) = P_N(N-k)$. We write $\mathbb{P}_+$ and $\mathbb{E}_+$ for probability and expectation given $X_1 = +1$. For even N we put $h = N/2$ and call $C_N = P_N(h)$ the center and $E_N = P_N(N) = P_N(0)$ the end. Whenever C_N appears, N is even. The crossing is the value of the parameter at which $C_N = E_N$. All logarithms are natural, $L = \log N$, $H_k = \sum_{j \leq k} 1/j$ is the k-th harmonic number, and γ is Euler's constant. Limits, $O(\cdot)$ and $o(\cdot)$ refer to $N \to \infty$, over even N when C_N appears. The constants implied by $O(\cdot)$ are absolute unless we name what they depend on.

The elephant walk (Definition 2.1) has the reversal probability $\varepsilon \in (0, 1)$ and $x = N\varepsilon$. Its crossing is ε_N, and $x_N = N\varepsilon_N$. The symbols that only one section uses are defined in that section, where they first appear.

1.2 Main results

We first state the approximation, then its consequences for the modes and the shape, and then the crossing.

The law $f_N(m) = \mathbb{P}(M_N = m)$ does not depend on X_1, and $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$ (Lemma 2.2). Let CP_N be the law of $\sum_{d=1}^{N-1} d\,\mathcal{N}_d$, where the $\mathcal{N}_d$ are independent Poisson variables with means $x/(d(d+1))$. Put $\lambda_m = (m - x\log x)/x$, and let

$$f_L(\lambda) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\lambda t} e^{B(t)}\,dt, \qquad B(t) = -it\log|t| - \frac{\pi}{2}|t|,$$

be the Landau density.

Theorem 1.1 (the approximation)**.** *For $N \geq 1$ and $0 < \varepsilon < 1$ the total variation distance between f_N and CP_N is at most an explicit $\delta_\Sigma = O(x^2 \log^2 N/N)$ (Theorem 3.8). Suppose that $x \to \infty$ and $x^3 \log^2 N/N \to 0$. Then, uniformly over integers $m \geq 0$,*

$$\bigl|x f_N(m) - f_L(\lambda_m)\bigr| = O\Bigl(\frac{1+\log x}{x} + \frac{x^3 \log^2 N}{N}\Bigr),$$

and $(M_N - x\log x)/x$ converges in law to the density f_L. Theorem 3.9 gives explicit finite-N bounds.

Theorem 1.2 (the modes)**.** *The Landau density has a unique mode $\lambda_{\mathrm{L}} \approx -0.222783$ (Lemma 3.10, computer-assisted). For $\log N \leq x \leq 2\log N$ and all sufficiently large N, every mode $a^* \geq N/2$ of P_N satisfies*

$$\bigl|N - a^* - x\log x - \lambda_{\mathrm{L}} x\bigr| \leq K\sqrt{x\log x},$$

and the same bound holds for every mode of f_N in place of $N - a^*$ (Corollary 3.11). We do not compute K.

Theorem 1.3 (the center and the crossing)**.** *Let $N \geq 2$ be even.*

(a) The ratio C_N/E_N is strictly increasing in $\varepsilon \in (0,1)$, and the crossing ε_N exists, is unique and lies in $(0, \frac{1}{2})$ (Proposition 2.3).

(b) For $0 < \varepsilon < \frac{1}{2}$ we have $C_N \geq \frac{4\varepsilon}{N+2}(1-2\varepsilon)$. If $\varepsilon(1+\log N) \leq \frac{1}{100}$, then $C_N \leq \frac{4\varepsilon}{N+2}\,N^{2\varepsilon}e^{5200\varepsilon}$ (Theorem 2.6, whose constant 5200 uses an enclosure in ball arithmetic).

(c) As $N \to \infty$ over even N, the crossing satisfies $x_N + \log x_N = 2\log N - \log 8 + O(\log^2 N/N)$, and so

$$x_N = 2\log N - \log\log N - \log 16 + O\Bigl(\frac{\log\log N}{\log N}\Bigr)$$

(Theorem 2.7).

So $C_N = \frac{4\varepsilon}{N+2}(1 + O(\varepsilon\log N))$ when $\varepsilon\log N$ is small, and the error factor is $1 + O(\log^2 N/N)$ at the crossing. For the persistent walk with switch probability q, the crossing has $T = Nq$ close to $\log N$. We compare the two crossings to the next order in the arXiv version of this paper [41].

At the crossing each hump is about $x\log x - 0.2228\,x$ steps from its end. For the shape between them put $D_N(a) = P_N(a+1) - 2P_N(a) + P_N(a-1)$ and $m^* = N - a^*$ for a mode $a^* \geq N/2$.

Theorem 1.4 (the shape)**.** *(a) For $N \geq 3$ and $0 < \varepsilon < \frac{1}{2}$, delete zeros from the signs of D_N and merge equal neighbors. The resulting pattern is a subsequence of $(+,-,+,-,+)$ (Theorem 3.16).*

(b) For every even $N \geq 8$, the crossing law has exactly two symmetric intervals of maximizing points. It strictly rises to each interval and strictly falls from it toward the center, its unique interior minimum (Corollary 3.19). This uses the finite endpoint-dip certificates of Proposition 3.7. The whole statement, certificates included, is kernel-checked in Lean (Section 5).

Between the humps we expect more, namely strict convexity on $[2m^*, N - 2m^*]$. The arXiv version proves this window for every large N, and at the crossing for $5 \leq N \leq 2000$ by a certified computation (the arXiv version of this paper [41]).

The shape of Theorem 1.4(b) needs no such window. It holds for every even $N \geq 8$. The computational inputs are described in Section 5, with the result each one supports. The tables and figures illustrate the theorems and are not used in their proofs.

Theorem 1.5 (the fixed-memory limit law)**.** *Fix $0 < \varepsilon < \frac{1}{4}$ and put $a = 1 - 2\varepsilon$. With a fair first step, $(2A_N - N)/N^a$ converges almost surely to a limit with an even density g_a. Either g_a is nonincreasing on $[0, \infty)$, or there is $c > 0$ such that g_a is nondecreasing on $[0, c]$ and nonincreasing on $[c, \infty)$ with $g_a(0) < g_a(c)$. So g_a has at most two modes (Theorem 4.2). The second case holds for $0 < \varepsilon \leq \frac{1}{80}$ (Theorem 4.3). With explicit rationals $z_- < z_+$ such that $z_- = 0.66881286001459775\ldots$ and $z_+ - z_- < 3.3 \cdot 10^{-15}$, the first case holds for $a \leq z_-$ and the second for $a \geq z_+$ (Theorem 4.4, computer-assisted).*

The proof of the threshold combines Taylor models in a with a bound on the characteristic function that improves itself through the fixed-point equation of Guérin, Laulin and Raschel (Remark 4.5).

Why twice as many reversals? An endpoint has mass about $\frac{1}{2}e^{-x}$. In the persistent walk, let q be the switch probability and $T = Nq$. Its runs have typical length N/T, and the right-step count spreads over about $N/\sqrt{T}$ values. Its central mass is of order $\sqrt{T}/N$, which balances e^{-T} at $T \approx \log N$. An elephant reversal instead starts a family of descendants, governed by Pólya's urn. Reaching the middle requires an early reversal whose family occupies about half the walk. This costs order ε, and the exact middle costs another $1/N$. Thus the center has mass about $4\varepsilon/N = 4x/N^2$, balancing the endpoint at $x \approx 2\log N$. Most of the roughly $2\log N$ reversals there occur too late to produce a large family.

1.3 Prior work

Fixed-parameter limits for the elephant walk are well developed [4, 5, 6]. In the superdiffusive regime the limit law has a smooth positive density with known moments [8, Theorems 1.3 and 1.4], and it is unimodal given the first step [7, Theorem 1.1]. Trinas and Valle prove a uniform local limit theorem [9, Theorem 3.1], and disprove a log-concavity conjecture of [7, Remark 2.3] with exact moments and a certified study of the recursion [9, Theorem 4.4]. They note that unimodality need not pass to the mixture given by a random first step. Our Theorem 4.2 treats this mixture for a fair first step, through the sign pattern at finite N. Engländer and Volkov's coin-turning walk instead switches at time n with probability p_n. For $p_n = a/n$ eventually, its occupation fraction tends to Beta(a, a) [16, Section 4, Theorem 1]. Fan and Qi [12] let the memory parameter of the elephant walk vary with time. We vary the reversal probability with the final length, not with time within one walk. So our walks form a triangular array, as do those of Roy, Takei and Tanemura [13], whose memory window varies with n.

Kürsten's percolation coupling [14] copies a parent with probability $1-2\varepsilon$ and otherwise gives the new vertex a fair sign. Baur and Bertoin [17] give the urn connection. Our regime is strongly supercritical in Baur's classification [15], following Bertoin [18]. These works describe cluster sizes. Baur's Corollary 1 gives a Poisson limit for the largest nonroot clusters. Miyazaki and Takei obtain related root-cluster limits via the minimal elephant walk [19, Theorems 2.3, 2.4 and Section 2.3]. They do not estimate the exact central mass in our regime.

Without truncation, our compound Poisson law is the Lea–Coulson Luria–Delbrück law. Its weak Landau limit as the mean grows is due to Pakes [23, p. 993]. Möhle proves it again with density formulas [24, Theorem 4.1 and Corollary 4.1], and credits it to Pakes. His Corollary 4.2 gives a uniform lattice local limit for integer parameters, without a rate. Kessler and Levine describe the small-parameter profile and the Landau limit [20, Eqs. (14), (18), (19)], as well as the dip threshold and mode near $\mu N(\log \mu N - 0.22)$ [20, Section 2]. On random recursive trees, Bertoin proves a weak limit of the same form for the number of vertices cut off from the root by rare cuts, when the cut probability is of order $k^{-1/4}$ on a tree of size k [26, Proposition 1]. His proof uses the coupling of Iksanov and Möhle [27, Section 2 and Proposition 1]. In it the subtrees removed by the cutting procedure of Meir and Moon [28] have the sizes of independent steps with law $1/(j(j+1))$, up to a first passage time. The same weights give the mean number $N/(d(d+1))$ of families of size d in Section 3. Schweinsberg [25] and Bertoin [26] obtain the same stable law for root-cluster fluctuations at retention $1-c/\log n$. What is new here is the total variation bound, the explicit local rate with truncation at $N-1$, the transfer to the elephant walk, and the shape theorems that follow.

Broadcasting papers study reconstruction of the root color from the terminal tree [2, 3]. Color-count limits and martingale error bounds are also available [29, 30]. Here the center is the probability of a color tie, relevant to majority-vote error, with the flip probability tending to zero. Gut and Stadtmüller's count [31, (1.5)] has next-step probability pk/n, whereas ours has $\varepsilon + (1-2\varepsilon)k/n$, so their chain cannot leave zero. The irreversible mutation expansion of Kessler and Levine [21, Eqs. (10), (13), (14)] has the same first derivative at $\varepsilon = 0$ as our law, but not our uniform relative bound at $N\varepsilon \asymp \log N$. We found no such central estimate in the literature reviewed.

1.4 Organization

Section 2 proves the crossing theorem, using the center bounds of Appendix A. Section 3 treats the bulk, modes and convexity. The compound Poisson and Landau proofs are in Appendix C. Appendix B states the auxiliary family estimates, Appendix D proves them, and Appendix E proves the second-difference lemmas. Section 4 treats the fixed-memory limit law, with the number of its modes. Appendix F proves the lemmas on sign patterns of functions used there, and Appendix G gives the certificate for the threshold. Section 5 describes the numerical and formal checks. The arXiv version of this paper [41] adds the center and the crossing to second order, the tail, further results for odd N and a convexity window between the humps. This article does not use them. That version also gives the full proofs of the steps in Appendix G. Every other proof used by the main theorems is in the article.

2 The elephant walk and its crossing

We reduce the elephant walk to a counting chain and prove that the crossing is unique. We then compute the center to first order in ε, bound it from both sides, and locate the crossing.

Definition 2.1 (elephant walk)**.** Fix $0 < \varepsilon < 1$. Let X_1 be uniform on $\{\pm 1\}$. For $t \geq 1$ let U_t be uniform on $\{1, \dots, t\}$, and put $X_{t+1} = X_{U_t}$ with probability $1 - \varepsilon$ and $X_{t+1} = -X_{U_t}$ with probability ε. All these choices are independent. We call the event $X_{t+1} = -X_{U_t}$ a flip. As in Section 1.1, $A_N = \#\{i \leq N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The crossing ε_N is the value of ε with $C_N = E_N$ (it exists and is unique by Proposition 2.3), and $x_N = N\varepsilon_N$.

A flip is what the introduction called a reversal. Given the first step, all that matters is how many steps disagree with it. So let M_t be the number of $i \leq t$ with $X_i \neq X_1$, and put

$$r_t(b) = \varepsilon + (1 - 2\varepsilon)\frac{b}{t}, \qquad f_N(m) = \mathbb{P}_+(M_N = m).$$

Lemma 2.2. *The process $(M_t)_{t \geq 1}$ is a Markov chain with $M_1 = 0$ and $\mathbb{P}(M_{t+1} = b + 1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$, and it is independent of X_1. For $0 \leq k \leq N$, $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$. In particular $C_N = f_N(h)$, $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$, and $P_N(k) = P_N(N-k)$.*

Proof. Given $X_1, \dots, X_t$, the copied step X_{U_t} differs from X_1 with probability M_t/t. So $X_{t+1} \neq X_1$ with conditional probability $(1-\varepsilon)M_t/t + \varepsilon(1 - M_t/t) = r_t(M_t)$. This depends only on M_t and not on X_1, which proves the first claim. On $\{X_1 = +1\}$ we have $A_N = N - M_N$, and on $\{X_1 = -1\}$ we have $A_N = M_N$. Averaging over X_1 gives the formula for $P_N(k)$, which does not change under $k \mapsto N - k$. Finally $M_N \leq N - 1$, so $f_N(N) = 0$, and $f_N(0) = \prod_{t=1}^{N-1}(1 - r_t(0)) = (1-\varepsilon)^{N-1}$. □

So the end is that of the persistent walk with switch probability ε. More flips always raise the center relative to the end.

Proposition 2.3. *Let $N \geq 2$ and $1 \leq k \leq N - 1$. Then $f_N(k)/f_N(0)$ and $P_N(k)/E_N$ are strictly increasing in $\varepsilon \in (0, 1)$. In particular C_N/E_N is strictly increasing, the crossing ε_N exists and is unique, and $\varepsilon_N < \frac{1}{2}$. Also $P_N(k) \geq 2E_N$ for $\varepsilon \geq \frac{1}{2}$, and in particular $C_N \geq 2E_N$ there.*

Proof. We compare each path of M with the path that never flips, one step at a time. The probability of a path of M is a product of $N - 1$ factors, namely $r_s(b_s)$ for an up step at time s and $1 - r_s(b_s)$ otherwise, where b_s is the state at time s. The path that never flips has probability $f_N(0) = (1-\varepsilon)^{N-1}$, which is one factor $1 - \varepsilon$ for each step. So we divide each factor by $1 - \varepsilon$. With $y = b_s/s \in [0, 1)$ the two kinds of factors become $\frac{\varepsilon + (1-2\varepsilon)y}{1-\varepsilon}$ and $1 - y\frac{1-2\varepsilon}{1-\varepsilon}$, whose derivatives in ε are $\frac{1-y}{(1-\varepsilon)^2} > 0$ and $\frac{y}{(1-\varepsilon)^2} \geq 0$. Both factors are positive, since $r_s(b_s) = \varepsilon(1-y) + (1-\varepsilon)y$ lies strictly between 0 and 1. So the ratio for each path is nondecreasing in ε, and it is strictly increasing if the path has an up step. Summing over the paths with $k \geq 1$ up steps proves the claim for $f_N(k)/f_N(0)$. By Lemma 2.2, $P_N(k)/E_N = [f_N(k) + f_N(N-k)]/f_N(0)$, and both k and $N - k$ are at least 1. So the claim for $P_N(k)/E_N$ follows.

It remains to locate the crossing. The ratio C_N/E_N is continuous and strictly increasing in ε. As $\varepsilon \to 0$ it tends to 0, since $C_N \to 0$ (at $\varepsilon = 0$ the chain never leaves 0) and $E_N \to \frac{1}{2}$. For $\varepsilon \geq \frac{1}{2}$ every $r_s(b)$ and every $1 - r_s(b)$ lies in $[1-\varepsilon, \varepsilon]$, so each path of M has probability at least $(1-\varepsilon)^{N-1} = 2E_N$. Every m with $0 \leq m \leq N - 1$ is the end of some path, so $f_N(m) \geq 2E_N$. Both k and $N - k$ lie in $\{1, \dots, N-1\}$, so $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)] \geq 2E_N$, and in particular $C_N \geq 2E_N$ there. So the ratio equals 1 at exactly one ε, and this ε lies in $(0, \frac{1}{2})$. □

2.1 First order, one early flip

When $\varepsilon = 0$ the chain M never leaves 0. After a first flip, and to first order in ε, it is Pólya's urn.

Lemma 2.4 (Pólya's urn)**.** *Let $2 \le v \le N$. At time v an urn holds one ball of one color and $v-1$ of the other. At each time $k = v, \dots, N-1$ we draw a ball uniformly and return it with one more ball of its color. Let $\beta_v(d)$ be the probability that at time N there are d balls of the first color. Then $\beta_v(d) = \binom{N-d-1}{v-2}/\binom{N-1}{v-1}$ for $1 \le d \le N-v+1$, and $\beta_v(d) = 0$ for other d. Also β_v is nonincreasing on $\{1, \dots, N-v+1\}$ and $\sum_d d\,\beta_v(d) = N/v$. For $1 \le d \le N-1$,*

$$\sum_{v=2}^{N} \beta_v(d) = \frac{N}{d(d+1)}, \qquad \sum_{v=2}^{N} (v-2)\,\beta_v(d) = \frac{2N(N-d-1)}{d(d+1)(d+2)}.$$

Proof. We compute the probability of one order of the draws and count the orders. At time k the urn holds k balls. So if s of them are of the first color, the next draw is of that color with probability s/k. Take one order of the $N-v$ draws in which i draws are of the first color. The draws of the first color contribute the numerators $1, 2, \dots, i$, the other draws contribute $v-1, v, \dots, N-i-2$, and the denominators are $v, v+1, \dots, N-1$. So this order has probability $\frac{i!\,(v-1)v\cdots(N-i-2)}{v(v+1)\cdots(N-1)} = \frac{(v-1)\,i!\,(N-i-2)!}{(N-1)!}$, whatever the order is. There are $\binom{N-v}{i}$ orders. With $d = i+1$ this gives $\beta_v(d) = a_d(v-1)\binom{N-v}{d-1}$ with $a_d = \frac{(d-1)!(N-d-1)!}{(N-1)!}$, and writing out the factorials shows that this is $\binom{N-d-1}{v-2}/\binom{N-1}{v-1}$. From the last form, $\beta_v(d)/\beta_v(d-1) = 1 - \frac{v-2}{N-d} \le 1$ for $2 \le d \le N-v+1$, so β_v is nonincreasing on its support. The fraction of balls of the first color is a martingale that starts at $1/v$. So the mean number of such balls at time N is N/v.

For the two sums we use the first form of $\beta_v(d)$ and put $i = N-v$, so that $v-1 = \binom{N-1-i}{1}$ and $(v-1)(v-2) = 2\binom{N-1-i}{2}$. The Chu–Vandermonde identity says that $\sum_i \binom{i}{a}\binom{N-1-i}{b} = \binom{N}{a+b+1}$ (sort the subsets of $\{1, \dots, N\}$ of size $a+b+1$ by their element number $a+1$, which we call $i+1$). With $a = d-1$ and $b = 1, 2$ it shows that the two sums are $a_d\binom{N}{d+1}$ and $2a_d\binom{N}{d+2}$, and these are the values in the statement. □

We need some more notation. Let $V_s(b)$ be the chance of ending at the center from state b at time s. We define it by the backward recursion $V_N(b) = \mathbf{1}\{b = h\}$ and $V_s(b) = r_s(b)V_{s+1}(b+1) + (1 - r_s(b))V_{s+1}(b)$ for $s < N$. So $V_s(b) = \mathbb{P}_+(M_N = h \mid M_s = b)$ whenever $\mathbb{P}_+(M_s = b) > 0$, and $V_s(b)$ is a polynomial in ε. In particular it is defined at $\varepsilon = 0$, without conditioning on an event of probability zero. For $1 \le t \le h$ let

$$\pi_t = \beta_{t+1}(h) = \frac{t}{N-t} \prod_{i=1}^{t-1} \frac{h-i}{N-i}, \qquad c_1 = \sum_{t=1}^{h} \pi_t, \qquad w_t = \frac{\pi_t}{c_1},$$

so that $(w_t)_{1 \le t \le h}$ is a probability vector.

The first-order coefficient in part (b) below is known. With both switching probabilities equal to ε and one sensitive cell at the start, the number of resistant cells in the reversible fluctuation assay of Bokes, Hlubinová and Singh [22, Section 2 and Eq. (1)] is our chain M. They condition on the first switch and use Pólya's urn, as we do, and their first-order formula [22, Eqs. (6)–(10)] gives $\varepsilon N/(m(m+1))$ as the part of $f_N(m)$ of first order in ε, which is $\frac{4\varepsilon}{N+2}$ at $m = h$. We include the short proof because the decomposition (a) and the weights (c) are what the uniform bounds of Theorem 2.6 and Appendix A start from.

Proposition 2.5 (first order, as in [22, Eqs. (6)–(10)])**.** *Let $N \ge 2$ be even.*

(a) The center is $C_N = \sum_{t=1}^{h} \varepsilon(1-\varepsilon)^{t-1} V_{t+1}(1)$.

(b) The center C_N is a polynomial in ε with $C_N(0) = 0$ (extended as a polynomial to $\varepsilon = 0$). At $\varepsilon = 0$ we have $V_{t+1}(1) = \pi_t$, and $C_N'(0) = c_1 = \frac{4}{N+2}$ exactly.

(c) The weights satisfy $\sum_t w_t t = 3 - \frac{6}{h+2}$*, and* $w_t \le 2t2^{-t}$ *for* $1 \le t \le h$*.*

Proof. (a) We condition on the time of the first flip. Under $\mathbb{P}_+$ the first flip happens at step $t+1$ exactly when $M_s = 0$ for $s \le t$ and $M_{t+1} = 1$. Since $r_s(0) = \varepsilon$, this has probability $(1-\varepsilon)^{t-1}\varepsilon$, and by the Markov property the chain then ends at h with probability $V_{t+1}(1)$. A path with no flip ends at $0 \ne h$. So C_N is the sum of $\varepsilon(1-\varepsilon)^{t-1}V_{t+1}(1)$ over $1 \le t \le N-1$. Only $t \le h$ contribute. Indeed at time $t+1$ there are t steps equal to X_1, this number never decreases, and at the center it is h. So $V_{t+1}(1) = 0$ for $t > h$.

(b) Every transition probability is affine in ε, so C_N and $V_{t+1}(1)$ are polynomials in ε. By (a), $C_N(0) = 0$ and $C_N'(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$. At $\varepsilon = 0$ we have $r_s(b) = b/s$. So from time $t+1$ on, M counts the balls of the first color in the urn of Lemma 2.4 with $v = t+1$, and $V_{t+1}(1) = \beta_{t+1}(h) = \pi_t$. Since $\beta_v(h) = 0$ for $v > h+1$, Lemma 2.4 with $d = h$ gives $c_1 = \sum_{v=2}^{N} \beta_v(h) = \frac{N}{h(h+1)} = \frac{4}{N+2}$.

(c) In the same way the second sum of Lemma 2.4, with $d = h$ and $v = t+1$, gives $\sum_t (t-1)\pi_t = \frac{2N(h-1)}{h(h+1)(h+2)}$. Dividing by $c_1 = \frac{N}{h(h+1)}$ gives $\sum_t w_t(t-1) = \frac{2(h-1)}{h+2} = 2 - \frac{6}{h+2}$, and adding $\sum_t w_t = 1$ gives the first claim. For the second claim, each factor $\frac{h-i}{N-i}$ in π_t is at most $\frac{1}{2}$, so $\pi_t \le \frac{t}{N-t}2^{1-t}$. Hence $w_t = \frac{N+2}{4}\pi_t \le \frac{N+2}{2(N-t)}\, t2^{-t}$. For $t \le h$ we have $N - t \ge h$ and $\frac{N+2}{2h} = 1 + \frac{2}{N} \le 2$, so $w_t \le 2t2^{-t}$. □

So to first order the walk flips once, at some step $t+1$, and then runs as Pólya's urn from t balls of one color and one of the other. As $N \to \infty$ the final fraction of the color that started with one ball tends in law to a Beta$(1,t)$ variable, whose density at $\frac{1}{2}$ is $t2^{1-t}$, and $\sum_t t2^{1-t} = 4$ explains the constant $4/(N+2)$. Since $w_t \le 2t2^{-t}$, only an early flip counts in this first-order term.

2.2 Two-sided bounds on the center

This subsection bounds the center from both sides. The bounds give the crossing in the next subsection.

Theorem 2.6. *Let* $N \ge 2$ *be even.*

(a) For $0 < \varepsilon < \frac{1}{2}$*,* $C_N \ge \frac{4\varepsilon}{N+2}(1-2\varepsilon)$*.*

(b) The bound $C_N \le \frac{4\varepsilon}{N+2}\, N^{2\varepsilon} e^{5200\varepsilon}$ *holds whenever* $0 < \varepsilon(1+\log N) \le \frac{1}{100}$*.*

In particular $\log(1-2\varepsilon) \le \log C_N - \log\frac{4\varepsilon}{N+2} \le \varepsilon(2\log N + 5200)$ *in the range of (b).*

Part (a) also holds for $\frac{1}{2} \le \varepsilon < 1$, trivially, because its right side is not positive there.

Idea of the proof. The proof is in Appendix A. After the first flip, M at time s is Pólya's urn with αs extra balls of each color, where $\alpha = \varepsilon/(1-2\varepsilon)$ (Lemma A.1). They pull the minority fraction toward $\frac{1}{2}$. For (a) we drop them. For (b) we compare M with an urn with $\kappa(s+10)$ extra balls, where $\kappa = 2\varepsilon/(1-4\varepsilon)$. One step of M at time s multiplies the hitting probability of this urn by a factor of at most about $e^{\kappa/s}$, and the product of these factors is about $e^{\kappa H_N} \approx N^{2\varepsilon}$.

Table 1: Elephant crossings x_N and the root g_N of $G(g) = \Lambda_N$. These values are numerical and are not used in any proof. We computed x_N in double precision from the recursion of Lemma 2.2, by Brent's method with an absolute tolerance of 10^{-12} in x, and we found g_N with a tolerance of about 10^{-15}.

N	x_N	g_N	$x_N/\log N$
50	3.940487	4.288636	1.007
400	7.629217	7.843768	1.273
1600	10.240565	10.340051	1.388
3200	11.546834	11.610464	1.431
12800	14.158845	14.182921	1.497
51200	16.778468	16.786946	1.547

2.3 The crossing

By Theorem 2.6, $C_N \approx 4x/N^2$ when $\varepsilon = x/N$ and x is of order $\log N$. Also $E_N = \frac{1}{2}(1-x/N)^{N-1} \approx \frac{1}{2}e^{-x}$. Setting them equal gives $x + \log x \approx 2\log N - \log 8$. So let

$$\Lambda_N = 2L - \log 8, \qquad G(x) = x + \log x,$$

and let g_N be the root of $G(g_N) = \Lambda_N$. It exists and is unique, since G increases from $-\infty$ to ∞ on $(0, \infty)$.

Theorem 2.7. *As $N \to \infty$ over even N, the following hold.*

(a) The crossing satisfies $x_N + \log x_N = \Lambda_N + O(L^2/N)$, so $x_N = g_N + O(L^2/N)$.

(b) The crossing satisfies $x_N = 2\log N - \log\log N - \log 16 + O(\log\log N/\log N)$.

Proof. (a) *Step 1. Both sides near the crossing.* Put $F(x) = \log C_N(x/N) - \log E_N(x/N)$ for $0 < x < N$, where $C_N(\varepsilon)$ and $E_N(\varepsilon)$ are the center and the end at flip probability ε. By Proposition 2.3, F is strictly increasing and x_N is its only zero. We show that

$$|F(x) - G(x) + \Lambda_N| \le \rho_N \quad (1 \le x \le 2\Lambda_N), \qquad \text{with } \rho_N = O(L^2/N). \tag{$*$}$$

Let $1 \le x \le 2\Lambda_N$ and $\varepsilon = x/N$. Then $\varepsilon \le 4L/N$, so for large N both parts of Theorem 2.6 apply, and together they say that $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon L)$. Here $\log \frac{4\varepsilon}{N+2} = \log x + \log 4 - 2L - \log(1 + 2/N)$, and $\log(1 + 2/N) \le 2/N \le 2xL/N$ because $x \ge 1$. For the end we use $\log(1-y) = -y + O(y^2)$ for $0 \le y \le \frac{1}{2}$ in $\log E_N = -\log 2 + (N-1)\log(1 - \frac{x}{N})$. So

$$\log C_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big), \qquad \log E_N = -\log 2 - x + O\Big(\frac{x^2}{N}\Big).$$

We subtract. Since $\log 4 + \log 2 = \log 8$, the main terms give $x + \log x - 2L + \log 8 = G(x) - \Lambda_N$, and both errors are $O(L^2/N)$ because $x \le 2\Lambda_N \le 4L$. This is ($*$).

Keeping the constants gives an explicit ρ_N. Suppose that $400L(1+L) \le N$. Then $\varepsilon(1+\log N) \le 4L(1+L)/N \le \frac{1}{100}$, so the inequality after part (b) of Theorem 2.6 applies. With $\log(1-y) \ge -y-y^2$ for $0 \le y \le \frac{1}{2}$ and $x \le 4L$, the error term for $\log C_N$ above lies between $-(3+8L)/N$ and $(8L^2 + 20800L)/N$, and the one for $\log E_N$ lies between $-16L^2/N$ and $4L/N$. So ($*$) holds with $\rho_N = (24L^2 + 20800L)/N$.

Step 2. Trap the zero. We have $G' = 1 + 1/x \geq 1$ and $G(g_N) = \Lambda_N$. So $G(x) - \Lambda_N < -\rho_N$ for $x < g_N - \rho_N$, and $G(x) - \Lambda_N > \rho_N$ for $x > g_N + \rho_N$. By $(*)$, $F(x) < 0$ for $1 \leq x < g_N - \rho_N$ and $F(x) > 0$ for $g_N + \rho_N < x \leq 2\Lambda_N$. Both ranges are nonempty for large N, because $\Lambda_N - \log \Lambda_N \leq g_N \leq \Lambda_N$ (note that $G(\Lambda_N - \log \Lambda_N) \leq \Lambda_N \leq G(\Lambda_N)$). Since F is increasing, its zero x_N lies between the two ranges, so $|x_N - g_N| \leq \rho_N$. Then x_N lies in $[1, 2\Lambda_N]$, and $F(x_N) = 0$ in $(*)$ gives $|G(x_N) - \Lambda_N| \leq \rho_N$. With the explicit ρ_N above, these two bounds hold for every even N with $L \geq 30$. Indeed $e^L \geq L^{12}/12!$ gives $24L^4 + 20800L^3 \leq N$, so $400L(1 + L) \leq N$ and $\rho_N \leq 1/L^2$, and both ranges are nonempty because $g_N \geq \Lambda_N - \log \Lambda_N > 50$.

(b) We solve $g_N = \Lambda_N - \log g_N$ by one substitution. Since $\Lambda_N - \log \Lambda_N \leq g_N \leq \Lambda_N$, the ratio g_N/Λ_N is $1 + O(\log \Lambda_N/\Lambda_N)$, so $\log g_N = \log \Lambda_N + O(\log \Lambda_N/\Lambda_N)$. Also $\Lambda_N = 2 \log N - \log 8$ gives $\log \Lambda_N = \log 2 + \log \log N + O(1/\log N)$. So $g_N = 2 \log N - \log 8 - \log 2 - \log \log N + O(\log \log N/\log N)$, which is (b) for g_N, and (a) gives it for x_N. □

Table 1 gives crossings for six values of N of the form $50 \cdot 2^k$. The limit 2 of $x_N/\log N$ is approached slowly, because of the term $-\log \log N$.

We prove three refinements in the arXiv version of this paper [41]. A lower bound for C_N matches Theorem 2.6(b) up to a factor $1 + O(\varepsilon)$. One more term of the crossing shows that $x_N < g_N$ for large N. A comparison with the crossing value z_N of T for the persistent walk shows that $x_N - (2z_N - \log(2\pi))$ is $c_*/\log N$ to first order, with $c_* = \frac{1}{2}\log(2\pi) - \frac{1}{4}$.

3 The shape of the law at the crossing

We now look at the whole law P_N near the crossing. In this section and in Appendices B and C, a statement that says "Assume (H)" assumes

$$N \geq 64, \qquad 0 < \varepsilon \leq \tfrac{1}{4}, \qquad x = N\varepsilon \geq 1. \tag{H}$$

Every other statement holds under its own hypotheses. Recall that $P_N(N - m) = \frac{1}{2}[f_N(m) + f_N(N - m)]$. Some statements also use odd N. There x_N is the value of x at which the two central values are as likely as the ends (it exists and is unique by Proposition 3.21). For a mode $a^* \geq N/2$ of P_N we put $m^* = N - a^*$, and m_h is the largest mode of f_N.

3.1 Families and the first flip

For $k_0 \geq 1$ let $(M_k^{[k_0]})_{k \geq k_0}$ be the Markov chain with $M_{k_0}^{[k_0]} = 0$ and the transition probabilities $r_k(b)$ of M. It is the count of disagreeing steps for a walk whose first k_0 steps are forced to agree. So $M^{[1]}$ has the law of M. We write $Y_d = M_d^{[1]}$, and $Z_t^{(j)} = M_t^{[j-1]}$ for $j \geq 2$ and $t \geq j - 1$. In a random recursive tree on $[N] = \{1, \dots, N\}$, vertex $k + 1$ is joined to a uniform vertex of $[k]$, independently for $k = 1, \dots, N - 1$. We write $\mathcal{T}_v$ for the subtree of v, that is, v and its descendants, and we call it the family of v.

Lemma 3.1 (tree and marks)**.** *Let $\mathcal{T}$ be the tree on $[N]$ in which the parent of $k + 1$ is U_k. For $2 \leq v \leq N$ let $\mu_v = 1$ if step v flips the step it copies, and $\mu_v = 0$ otherwise. Then $\mathcal{T}$ is a random recursive tree, the marks μ_v are i.i.d. Bernoulli(ε) and independent of $\mathcal{T}$, and $X_v = X_1(-1)^{\sigma(v)}$, where $\sigma(v)$ is the number of marks on the path from v to the root, the root excluded. So $M_N = \#\{v : \sigma(v) \text{ odd}\}$. If the marks on $2, \dots, k_0$ are set to 0, then the number of odd vertices in $[k]$, $k \geq k_0$, is a copy of the chain $M^{[k_0]}$.*

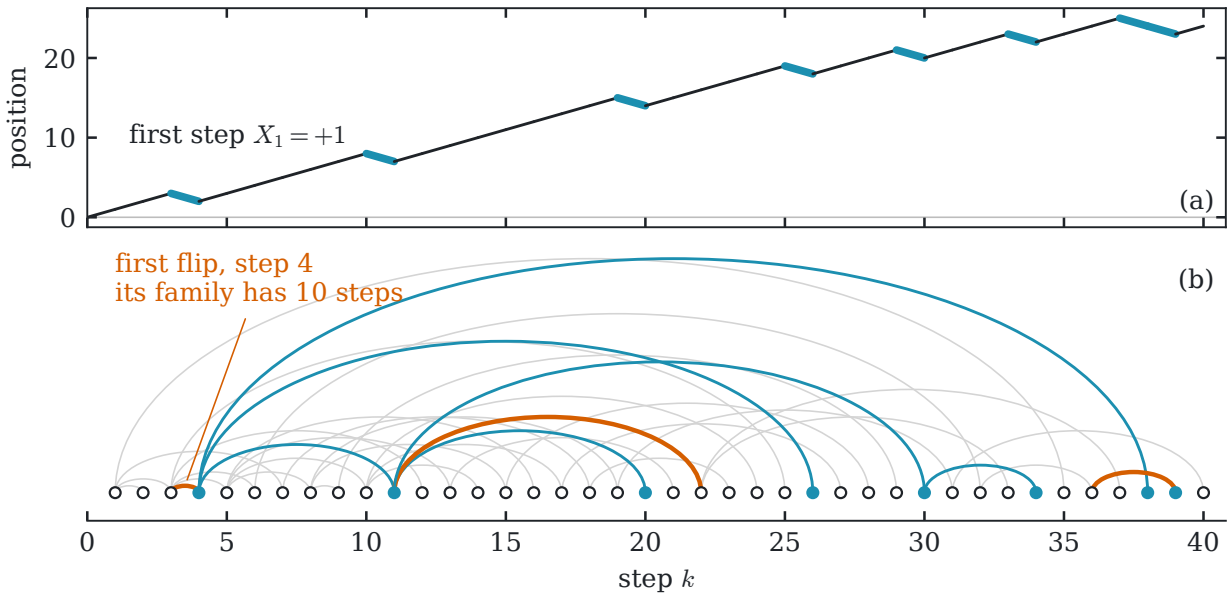


Figure 1: One elephant walk with $N = 40$ steps and $\varepsilon = 0.07$. (a) The position against the step. The steps that disagree with the first step are drawn heavier and in blue. (b) The memory tree of the same walk. Step $k+1$ copies the step at the left end of its arc, and the arcs drawn heavier are flips. A step disagrees with the first step exactly when the path from it to step 1 crosses an odd number of flips (Lemma 3.1). The first flip, at step 4, turns its family of ten steps. The flip at step 22, inside that family, turns its own family of three steps back, and the flip at step 39 turns only that step.

Proof. The parent U_k of $k+1$ is uniform on $[k]$ and independent of the earlier choices, and the marks are the flip indicators. The path from $k+1$ to the root is $k+1$ followed by the path from U_k, so $\sigma(k+1) = \mu_{k+1} + \sigma(U_k)$ and, by induction on k, $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$. Now set $\mu_2 = \cdots = \mu_{k_0} = 0$. If b of the vertices in $[k]$ are odd, $k \geq k_0$, then $k+1$ is odd with conditional probability $(1-\varepsilon)\frac{b}{k} + \varepsilon(1 - \frac{b}{k}) = r_k(b)$, whatever else is known about $[k]$. □

So a single flip at v turns every step of its family (Figure 1). For $k \geq v$, if $\mathcal{T}_v$ contains s of the vertices in $[k]$, then $k+1$ joins $\mathcal{T}_v$ with probability s/k. This is the urn of Lemma 2.4, so $|\mathcal{T}_v|$ has the law β_v and mean N/v, and on average there are $N/(d(d+1))$ families of size d. If v is the only marked vertex, the odd vertices are those of $\mathcal{T}_v$. Two or more marks have probability $O(\varepsilon^2)$ at fixed N, and no mark gives $M_N = 0$. So at fixed N the part of $f_N(m)$ of first order in ε is exactly $\varepsilon \sum_v \beta_v(m) = x/(m(m+1))$ for $1 \leq m \leq N-1$, which is the formula of [22, Eq. (10)] with one cell at the start. It is also proved in Lean for the chain M (Section 5).

To go beyond first order we cut the tree at the first flip.

Lemma 3.2 (splitting)**.** *Let $\mathcal{T}$ be a random recursive tree on $[N]$, let $2 \leq v \leq N$ and let $S \subseteq [N]$ with $\min S = v$. Given that $\mathcal{T}_v$ has vertex set S, the restrictions of $\mathcal{T}$ to S and to $[N] \setminus S$, relabeled in increasing order, are independent random recursive trees. They are independent of the parent of v, which is uniform on $[v-1]$.*

Proof. The tree $\mathcal{T}$ is uniform over the $(N-1)!$ increasing trees on $[N]$. Its family $\mathcal{T}_v$ has vertex set S exactly when every $u \in S \setminus \{v\}$ has its parent in S and every $u \notin S$ with $u \geq 2$ has its parent outside S. So, given this event, each element of S other than v picks its parent freely among the

smaller elements of S, v picks any element of $[v-1]$, and each element of $[N] \setminus S$ other than 1 picks freely among the smaller elements of $[N] \setminus S$. This is the claim. □

Lemma 3.3 (the first flip)**.** *Let $V = \min\{v \geq 2 : \mu_v = 1\}$, with $V = \infty$ and $|\mathcal{T}_V| = 0$ if there is no mark. If $V < \infty$, let $\bar{Y}$ be the number of vertices u of $\mathcal{T}_V$ with $\sigma(u)$ even, and let $\bar{Z}$ be the number of odd vertices outside $\mathcal{T}_V$. Then for $2 \leq j \leq N$, $1 \leq d \leq N-j+1$ and integers y, z,*

$$\mathbb{P}\big(V = j,\, |\mathcal{T}_V| = d,\, \bar{Y} = y,\, \bar{Z} = z\big) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)\,\mathbb{P}(Y_d = y)\,\mathbb{P}\big(Z^{(j)}_{N-d} = z\big).$$

In particular $\mathbb{P}(V = j,\, |\mathcal{T}_V| = d) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and for $m \geq 1$

$$f_N(m) = \sum_{j=2}^{N} \varepsilon(1-\varepsilon)^{j-2} \sum_{d=1}^{N-j+1} \beta_j(d)\,\mathbb{P}\big(d - Y_d + Z^{(j)}_{N-d} = m\big),$$

where Y_d and $Z^{(j)}_{N-d}$ are independent.

Proof. Marks and tree are independent, and $|\mathcal{T}_j|$ has the law β_j, which gives the law of $(V, |\mathcal{T}_V|)$. Condition on $V = j$ and on the vertex set S of $\mathcal{T}_j$, with $|S| = d$. The marks on $j+1, \ldots, N$ are still i.i.d., and by Lemma 3.2 the trees on S and on $[N] \setminus S$ are independent random recursive trees. The vertices $1, \ldots, j-1$ carry no mark, so j is odd, and $u \in S$ is even exactly when the path from u up to j, with j excluded, has an odd number of marks. So by Lemma 3.1 for the tree on S rooted at j, the count $\bar{Y}$ has the law of Y_d. Paths from outside S stay outside S, and the tree on $[N] \setminus S$ has the unmarked vertices $1, \ldots, j-1$ first. So by Lemma 3.1 the count $\bar{Z}$ has the law of $Z^{(j)}_{N-d}$. The two counts are functions of the two trees and of the marks on the two disjoint sets S and $[N] \setminus S$, so they are independent, and their joint law depends on S only through d. This gives the joint law. On $\{V = j\}$ we have $M_N = d - \bar{Y} + \bar{Z}$, and $M_N = 0$ if there is no mark. Summing over j, d and the pairs (y, z) with $d - y + z = m$ gives the formula for $f_N(m)$. □

So the first flip turns its whole family except the steps of that family that flip back, and the rest of the tree is a walk whose first $j-1$ steps agree with X_1. Only the law of each pair $(Y_d, Z^{(j)}_{N-d})$ enters the lemma. So from now on $(Y_d)_{d \geq 1}$ is a copy of the chain $M^{[1]}$, and $(Z^{(j)}_t)_{t \geq j-1}$ is a copy of $M^{[j-1]}$, independent of Y and of $(V, |\mathcal{T}_V|)$.

Lemma 3.4 (unimodality given the first step [7, Section 2, proof of Theorem 1.1])**.** *The function f_N is unimodal.*

This is shown at the start of the proof of Theorem 1.1 of [7], for every N and every memory parameter $p = 1 - \varepsilon \in [0, 1]$, for the position $N - 2M_N$ given $X_1 = +1$. The lemma is also proved in Lean (Section 5). So f_N is nondecreasing on $\{0, \ldots, m_h\}$ and nonincreasing on $\{m_h, \ldots, N\}$. With a fair first step, P_N mixes f_N and its mirror image. When the mode of f_N is far from the center, the mixture can have two humps, one near each end.

3.2 The end and its neighbor

We first say heuristically why the end should fall below its neighbor once $x > 2$ (Figure 2(a), inset). Given $X_1 = +1$, the end needs no flip at all. Its neighbor, the value $N-1$, needs one flip that is never copied. A flip at step j is never copied with probability about j/N. Summing $\varepsilon j/N$ over j gives a factor of about $x/2$ in favor of the neighbor.

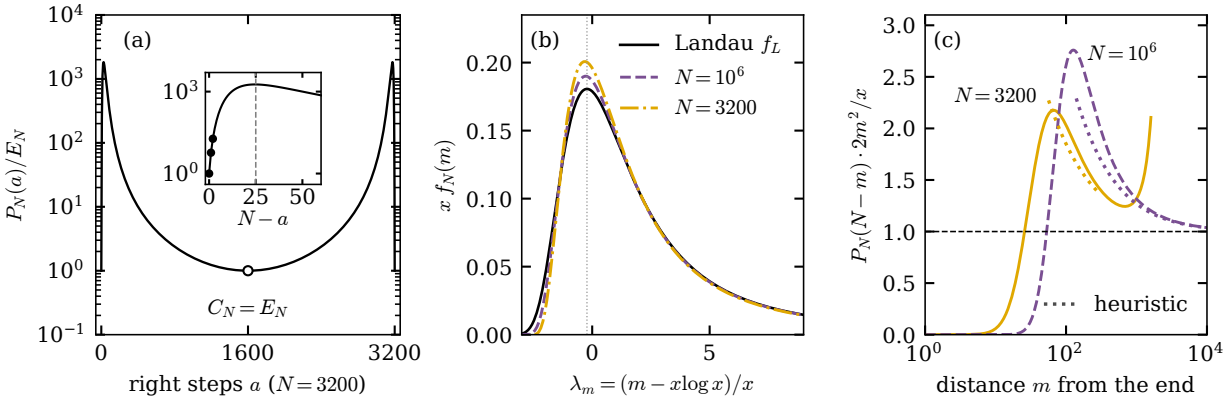


Figure 2: The elephant walk at its crossing. (a) The law P_N for $N = 3200$ at its crossing $x = 11.5468$, divided by the end E_N. The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 values, with dots at N, $N-1$ and $N-2$. The end is below its neighbor, and the modes are at distance 25 from each end (dashed). (b) The law of M_N, scaled as $x f_N(m)$, against $\lambda_m = (m - x\log x)/x$, with the Landau density f_L. The dotted line is its mode λ_{L}. (c) The tail $P_N(N-m)$ divided by $x/(2m^2)$. The dashed line is the limit 1 of the tail theorem in the arXiv version of this paper [41], and the dotted curves are the heuristic correction $1 + x(2\log m + 2\gamma - 3)/m$ for $2x\log x \le m \le N/x$. The rise of the $N = 3200$ curve near $m = N/2$ is the mirror term $f_N(N-m)$. At $N = 10^6$ we use $x = g_N = 22.4407$.

Theorem 3.5 (endpoint dip)**.** *Let $N \ge 2$ and $0 < \varepsilon < 1$, and put $\rho_1 = P_N(N-1)/E_N$. Then*

$$\rho_1 > \frac{x}{2(1-\varepsilon)}.$$

If moreover $N \ge 3$ and $\varepsilon \le \frac{1}{2}$, then

$$\rho_1 \le \frac{x(1+1/N)^2}{2-3\varepsilon} + \frac{\varepsilon}{1-\varepsilon}.$$

Hence the end is strictly below its neighbor if $x \ge 2(1-\varepsilon)$. If $N \ge 3$, $\varepsilon \le \frac{1}{2}$ and the right side of the second bound is below 1, the end is strictly above its neighbor.

Proof. By Lemma 2.2, $\rho_1 = \big(f_N(1) + f_N(N-1)\big)/f_N(0)$ with $f_N(0) = (1-\varepsilon)^{N-1}$. We bound the two terms in turn.

The term $f_N(1)$. On the event $M_N = 1$ the chain moves up exactly once, say from time $j-1$ to time j with $2 \le j \le N$. Before that it stays at 0 for $j-2$ steps, each with probability $1 - r_t(0) = 1 - \varepsilon$. The move up has probability ε. After it the chain stays at 1 at the times $k = j, \dots, N-1$, each with probability $1 - r_k(1) = (1-\varepsilon)(1 - c'/k)$, where $c' = (1-2\varepsilon)/(1-\varepsilon) \le 1$. Multiplying and summing over j gives

$$f_N(1) = \varepsilon(1-\varepsilon)^{N-2} \sum_{j=2}^{N} \vartheta_j, \qquad \vartheta_j = \prod_{k=j}^{N-1} \Big(1 - \frac{c'}{k}\Big).$$

For the lower bound we use $c' \le 1$. Then $\vartheta_j \ge \prod_{k=j}^{N-1}(1 - 1/k) = (j-1)/(N-1)$, and these lower bounds sum to $N/2$ over $2 \le j \le N$. So $f_N(1)/f_N(0) \ge x/(2(1-\varepsilon))$. For the upper bound

let $N \geq 3$ and $\varepsilon \leq \frac{1}{2}$, so that $c' \geq 0$. Then $1 - c'/k \leq e^{-c'/k}$ and $\sum_{k=j}^{N-1} 1/k \geq \log(N/j)$ give $\vartheta_j \leq (j/N)^{c'}$. As $(t/N)^{c'}$ is nondecreasing in t, we can compare the sum with an integral, namely

$$\sum_{j=2}^{N} \vartheta_j \leq \int_2^{N+1} \Big(\frac{t}{N}\Big)^{c'} dt \leq \frac{N(1+1/N)^{c'+1}}{c'+1} \leq \frac{N(1+1/N)^2(1-\varepsilon)}{2-3\varepsilon},$$

using $c' \leq 1$ and $1/(c'+1) = (1-\varepsilon)/(2-3\varepsilon)$. So $f_N(1)/f_N(0) \leq x(1+1/N)^2/(2-3\varepsilon)$.

The term $f_N(N-1)$. The event $M_N = N-1$ needs an up move at every time. The first has probability $r_1(0) = \varepsilon$, and the one at time $k \geq 2$ has probability $r_k(k-1) = 1 - \varepsilon - (1-2\varepsilon)/k$. This lies strictly between 0 and 1, as every $r_k(b)$ with $0 < b < k$ does, and it lies in $[\frac{1}{2}, 1-\varepsilon]$ when $\varepsilon \leq \frac{1}{2}$. So $f_N(N-1) > 0$, and $f_N(N-1) \leq \varepsilon(1-\varepsilon)^{N-2}$ when $\varepsilon \leq \frac{1}{2}$. This makes the lower bound strict and gives the term $\varepsilon/(1-\varepsilon)$ in the upper bound. (For $N = 2$ the values 1 and $N-1$ coincide, and the argument is the same.) □

Corollary 3.6 (the dip at the crossing)**.** *Let $N \geq 2$ be even and $0 < \varepsilon < 1$, and suppose that $x \geq 2(1-\varepsilon)$ and $C_N \leq E_N$. Then $C_N \leq E_N < P_N(N-1)$, and P_N is not unimodal. In particular, if $x_N \geq 2(1-\varepsilon_N)$, then at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal.*

Proof. Theorem 3.5 gives $P_N(N-1) > E_N \geq C_N$. Suppose that P_N is nondecreasing on $\{0, \dots, a_0\}$ and nonincreasing on $\{a_0, \dots, N\}$. It is symmetric about h, so we may replace a_0 by $N - a_0$ and assume $a_0 \leq h$. Then $P_N(a) \leq C_N$ for every $a \geq h$, including $a = N-1$, which is a contradiction. □

By Proposition 2.3, $C_N \leq E_N$ holds exactly when $\varepsilon \leq \varepsilon_N$, and $x \geq 2(1-\varepsilon)$ says that $\varepsilon \geq 2/(N+2)$. So the corollary covers the ε with $2/(N+2) \leq \varepsilon \leq \varepsilon_N$. In particular the condition $x_N \geq 2(1-\varepsilon_N)$ says that $\varepsilon_N \geq 2/(N+2)$. By Proposition 2.3 it holds exactly when $C_N \leq E_N$ at $\varepsilon = 2/(N+2)$. We now show that this is so for every even $N \geq 8$.

Proposition 3.7 (the hypothesis holds from $N = 8$ on, computer-assisted)**.** *For every even $N \geq 8$ we have $C_N < E_N$ at $\varepsilon = 2/(N+2)$, and hence $x_N > 2(1-\varepsilon_N)$. So for every even $N \geq 8$, at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal. For $N = 2$, 4 and 6 the condition $x_N \geq 2(1-\varepsilon_N)$ fails, and no ε meets the hypotheses of Corollary 3.6.*

Proof. The range $N \geq 1688$. Take $\varepsilon = 2/N$. Then $\varepsilon(1 + \log N) \leq \frac{1}{100}$, so Theorem 2.6(b) gives $C_N \leq \frac{8}{N(N+2)} N^{4/N} e^{10400/N}$. The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1688$. Since $\log(1-y) \geq -y - y^2$ for $0 \leq y \leq \frac{1}{2}$, we also have $E_N = \frac{1}{2}(1 - \frac{2}{N})^{N-1} \geq \frac{1}{2} e^{-2-2/N} > 0.06$. So $C_N < E_N$ at $\varepsilon = 2/N$, and Proposition 2.3 gives $\varepsilon_N > 2/N$. Hence $C_N < E_N$ at $\varepsilon = 2/(N+2)$ as well.

The range $8 \leq N < 1688$. For $N = 8$ the ratio C_N/E_N at $\varepsilon = 2/(N+2)$ is exactly $\frac{67903}{71680} < 1$. For even $10 \leq N \leq 2000$ the script `certify_center.py` encloses ε_N in an interval, by computing $C_N - E_N$ in ball arithmetic at its two ends and using Proposition 2.3, and it checks by an exact rational comparison that the lower end of the enclosure lies above $2/(N+2)$. So $\varepsilon_N > 2/(N+2)$, and Proposition 2.3 gives $C_N < E_N$ at $\varepsilon = 2/(N+2)$. The ratio of the lower end to $2/(N+2)$ is at least 1.107, with its minimum at $N = 10$.

The three smallest N. At $\varepsilon = 2/(N+2)$ the ratio C_N/E_N is exactly 2, $\frac{5}{3}$ and $\frac{508}{405}$ for $N = 2$, 4 and 6, so $\varepsilon_N < 2/(N+2)$ there. These ratios and the one for $N = 8$ are computed in exact rational arithmetic by `certify_elephant.py`. The last claim of the proposition about $N \geq 8$ is Corollary 3.6. □

3.3 The bulk

The steps that disagree with X_1 lie in the flipped families. Recall the compound Poisson law CP_N of Section 1.2. On average $x/(d(d+1))$ flipped vertices have a family of size d. If no flipped step has a flipped descendant, which is likely when $x^2 \log N/N$ is small, then M_N is the total size of the flipped families, and this is close to the compound Poisson sum.

The CP law and its Landau limit are known (Section 1.3). We give an explicit total variation bound for the CP approximation to the elephant law, and an explicit local rate for the law with jump sizes truncated at $N-1$. Numerical values of f_L and its mode are standard (see Kölbig and Schorr [32]). Lemma 3.10 adds a certified enclosure.

Theorem 3.8 (compound Poisson bulk)**.** *For $N \geq 1$ and $0 < \varepsilon < 1$,*

$$\mathrm{TV}(f_N, \mathrm{CP}_N) \leq \delta_\Sigma := \varepsilon^2(NH_N - 2N + 1) + 2\varepsilon^2(N-1) + 4\varepsilon^2 \sum_{k=0}^{N-2} (H_k^2 + H_k).$$

Also $\delta_\Sigma \leq \delta_B := \frac{x^2 \log N}{N} + \frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}$.

Here $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_k |P(k) - Q(k)|$. Thus the same bound holds for each mass and each event. Also $\delta_\Sigma = O(x^2 \log^2 N/N)$. Appendix C.1 proves the bound by coupling marks with Poisson variables and revealing the tree's parents one at a time. Changing one parent affects family sizes only along two paths to the root, whose depths have second moments of order $\log^2 N$.

The families of size up to about x add up to nearly $x \log x$, and the few families of size of order x give fluctuations of order x. So we center M_N at $x \log x$ and divide by x, which gives the coordinate λ_m and the Landau density f_L of Section 1.2. It is a stable density with index one. We show in Appendix C that e^B is the characteristic function of a probability law of class L, so that f_L is a unimodal probability density by [33].

Theorem 3.9 (Landau limit)**.** *(a) Let $x \geq 1$ and $N \geq 4x$. For every integer m, with $\mathrm{CP}_N(m) = 0$ for $m < 0$,*

$$\big|x\,\mathrm{CP}_N(m) - f_L(\lambda_m)\big| \leq E_0 := \frac{889 + 165\log x}{x} + \frac{6.6\,x}{N} \leq 889\Big(\frac{1+\log x}{x} + \frac{x}{N}\Big).$$

(b) Let $0 < \varepsilon \leq \frac{1}{4}$ and $x \geq 1$. For every integer $m \geq 0$,

$$\big|x f_N(m) - f_L(\lambda_m)\big| \leq E_0 + x\delta_\Sigma \leq 889\Big(\frac{1+\log x}{x} + \frac{x}{N}\Big) + x\delta_B.$$

(c) Let $N_n \geq 1$ and $0 < \varepsilon_n < 1$ for $n \geq 1$, and put $x_n = N_n\varepsilon_n$. If $x_n \to \infty$ and $x_n^3 \log^2 N_n/N_n \to 0$, then $(M_{N_n} - x_n \log x_n)/x_n$ converges in law to the law with density f_L.

Near the crossing the number of disagreeing steps is about $x \log x$, with fluctuations of order x that follow the Landau law, point by point (Figure 2(b)). Both conditions of (c) hold at the crossing, by Theorem 2.7. For the untruncated law the weak limit in (c) is Pakes' theorem [23, p. 993], and Bertoin proves the analogue for the vertices cut off from the root of a random recursive tree in another regime [26, Proposition 1]. So (c) is new only through the transfer to the walk. We know of no earlier form of (a) and (b).

Appendix C proves the local bound by expanding the compound Poisson exponent at zero and applying Fourier inversion. The expansion gives the centering $x \log x$, and its remainder gives the rate.

To locate the modes we need to know that f_L is curved at its maximum.

Lemma 3.10 (the mode of the Landau density, computer-assisted)**.** *(a) The density f_L has exactly one mode λ_{L}, and $-0.22278299 < \lambda_{\mathrm{L}} < -0.22278297$. It is nondecreasing on $(-\infty, \lambda_{\mathrm{L}}]$ and nonincreasing on $[\lambda_{\mathrm{L}}, \infty)$.*

(b) The derivative vanishes at the mode, $f_L'(\lambda_{\mathrm{L}}) = 0$, and $-0.0792 \le f_L''(\lambda) \le -0.0786$ for $-0.2229 \le \lambda \le -0.2227$.

(c) For every real λ, $f_L(\lambda_{\mathrm{L}}) - f_L(\lambda) \ge \kappa_L \min\{1, (\lambda - \lambda_{\mathrm{L}})^2\}$ with $\kappa_L = 2.6 \cdot 10^{-10}$.

The proof is in Appendix C.3, and it is computer-assisted. Lean proves the lemma as well, with the unimodality of f_L that [33] gives as its only input (Section 5), so the computation in ball arithmetic is an independent check.

Corollary 3.11 (where the modes are)**.** *(a) Assume (H). Let κ_L be as in Lemma 3.10(c), let $E_1 = 889\big(\frac{1+\log x}{x} + \frac{x}{N}\big) + x\delta_B$ be the bound of Theorem 3.9(b), and put*

$$E' = E_1 + \frac{16x^2 H_N}{N^2} + \frac{\|f_L''\|_\infty}{8x^2}.$$

If $E' < \kappa_L/2$ and $m_h \le N/4$, then every mode $a^* \ge N/2$ of P_N satisfies $|m^* - x\log x - \lambda_{\mathrm{L}} x| \le x\sqrt{2E'/\kappa_L}$, and so does every mode of f_N.

(b) There are K and N_2 such that the following holds for every $N \ge N_2$, even or odd, and every ε with $\log N \le x \le 2\log N$. Every mode $a^* \ge N/2$ of P_N has

$$\big|m^* - x\log x - \lambda_{\mathrm{L}} x\big| \le K\sqrt{x\log x},$$

and so has every mode of f_N in place of m^*. In particular this holds at the crossing, with $x = x_N$, for every large even N (Theorem 2.7(b)).

In (a) every constant is explicit. (Here $\|f_L''\|_\infty \le 16/\pi^4$, since $|e^{B(t)}| = e^{-\pi|t|/2}$.) We do not compute K and N_2 in (b).

So each hump of the law at the crossing sits about $x\log x - 0.2228\,x$ steps from its end. The proof is in Appendix C.4. It compares f_N at a mode with f_N at the integer nearest to $x\log x + \lambda_{\mathrm{L}} x$, and uses Theorem 3.9(b) and Lemma 3.10(c).

Remark 3.12 (how good the bulk approximation is). The bounds of this subsection are far from sharp at the sizes we can compute. At $N = 10^6$ and $x = 22.4407$ the bound δ_B is still about 0.46, but there the computed law has $|f_N(m)/\mathrm{CP}_N(m) - 1| \le 3.6 \cdot 10^{-4}$ for $1 \le m \le 10^4$. At every tested crossing m^* equals the mode of CP_N. The proof of Corollary 3.11(b) needs the bound of Theorem 3.9(b) to be below $\kappa_L/2$, where $\kappa_L = 2.6 \cdot 10^{-10}$ is the constant of Lemma 3.10(c). With the constant 889 this needs $x > 2 \cdot 10^{14}$, and numerically the best κ_L is about 0.026. These numbers are computed in double precision and are not used in any proof.

3.4 The tail

Beyond the humps the law falls like $x/(2m^2)$. We prove in the arXiv version of this paper [41] that $P_N(N-m) = \frac{x}{2m^2}(1 + o(1))$ for even N with $x \ge 1$ and $x\log^2 N/N \to 0$, uniformly for $\omega_N x \log^2(x+1) \le m \le \delta_N' N$, for every $\omega_N \to \infty$ and every $\delta_N' \to 0$.

At $N = 10^6$, $x = 22.4407$ and $m = 10^4$ the computed ratio $P_N(N-m)/(x/(2m^2))$ is 1.03777. Heuristically, at moderate m the law CP_N suggests $f_N(m)\,m(m+1)/x \approx 1 + x(2\log m + 2\gamma - 3)/m$, which we do not prove. This factor is still about 2 near $m = 70$ when $N = 3200$ (Figure 2(c)).

3.5 The shape at every even crossing

This subsection proves that the second difference of P_N changes sign at most four times. At every even crossing with $N \geq 8$ this gives the unique interior minimum and the two intervals of modes, with no asymptotic input.

Throughout this subsection and the next, $0 < \varepsilon < \frac{1}{2}$ and $c = 1 - 2\varepsilon$, so that $r_t(b) = \varepsilon + cb/t$. For $t \geq 1$ and every integer a we write $P_t(a) = \mathbb{P}(A_t = a)$, which is 0 unless $0 \leq a \leq t$, and

$$D_t(a) = P_t(a+1) - 2P_t(a) + P_t(a-1).$$

We also put $s_t = D_t(1) = P_t(0) - 2P_t(1) + P_t(2)$ for $t \geq 2$.

Lemma 3.13 (recurrence for the second difference)**.** *For every $t \geq 1$ and every integer a,*

$$P_{t+1}(a) = \big(1 - r_t(a)\big)P_t(a) + r_t(a-1)\,P_t(a-1), \tag{1}$$
$$D_{t+1}(a) = \big(1 - r_t(a+1)\big)D_t(a) + r_t(a-2)\,D_t(a-1). \tag{2}$$

Also $P_t(a) = P_t(t-a)$ and $D_t(a) = D_t(t-a)$. The same two recurrences hold with $f_t(m)$ in place of $P_t(a)$.

Proof. Given $A_t = a$, the copied step is $+1$ with probability a/t, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{a}{t} + \varepsilon(1 - \frac{a}{t}) = r_t(a)$. This gives (1) for $0 \leq a \leq t+1$, where at $a = 0$ and at $a = t+1$ the formula has one extra term, namely $r_t(-1)P_t(-1)$ or $(1 - r_t(t+1))P_t(t+1)$, and this term is 0 because $P_t(-1) = P_t(t+1) = 0$. For other a every term vanishes. The symmetry of P_t is Lemma 2.2, and it gives the symmetry of D_t. The chain M of Lemma 2.2 has the same transition probabilities, so (1) holds for f_t as well.

For (2) put $\varsigma = c/t$, so that $r_t(b+1) = r_t(b) + \varsigma$. We insert (1) at $a+1$, a and $a-1$ into the definition of $D_{t+1}(a)$ and collect the coefficients of $P_t(a+1)$, $P_t(a)$, $P_t(a-1)$ and $P_t(a-2)$. They are

$$1 - r_t(a+1), \qquad 3r_t(a) - 2, \qquad 1 - 3r_t(a-1), \qquad r_t(a-2).$$

The right side of (2) has the coefficients $1-r_t(a+1)$, $-2+2r_t(a+1)+r_t(a-2)$, $1-r_t(a+1)-2r_t(a-2)$ and $r_t(a-2)$. The second ones agree because $2r_t(a+1) + r_t(a-2) = 3r_t(a)$, and the third ones agree because $r_t(a+1) + 2r_t(a-2) = 3r_t(a-1)$. □

Trinas and Valle [9, (2.2)] write (1) for the mass function of the position, which is the forward equation of the counting chain of Lemma 2.2, and they use it for a local limit theorem in the superdiffusive regime. In [9, (4.10)] they write the analogue of (2) for differences of higher order, and they use it at a fixed memory parameter in a certified study of log-concavity. We use (2) for signs. The coefficients in (2) are positive when $2 \leq a \leq t-1$. So the second difference moves like a mass that is partly killed, and its sign can change only through what enters at the two edges. The next lemma is the elementary form of the variation diminishing property that we need.

For a finite list z of real numbers let $\mathrm{sp}(z)$ be its sign pattern. We delete the zeros of z, replace each entry by its sign, and merge equal neighbors. So $\mathrm{sp}(z)$ is a list of alternating signs. For two such lists we write $p \preceq q$ if p is a subsequence of q. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

Lemma 3.14 (one step)**.** *Let $\zeta = (\zeta_1, \dots, \zeta_n)$ be real, and let $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ for $1 \leq i \leq n-1$ with $A_i \geq 0$ and $B_i \geq 0$. Let $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. Then $\mathrm{sp}(w) = \mathrm{sp}(\zeta)$ and $\mathrm{sp}(\eta) \preceq \mathrm{sp}(\zeta)$.*

We prove this in Appendix E.1.

Lemma 3.15 (the sequence next to the end)**.** *For every $t_1 \geq 2$ we have $\mathrm{sp}(s_2, s_3, \ldots, s_{t_1}) \preceq \Pi_3$. Also, for every $0 \leq \varepsilon \leq 1$, the numbers s_3 and s_4 do not both vanish.*

We prove this in Appendix E.2.

Theorem 3.16 (at most five blocks)**.** *Let $N \geq 3$ and $0 < \varepsilon < \frac{1}{2}$. Then*

$$\mathrm{sp}\big(D_N(1), D_N(2), \ldots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

So the second difference of P_N changes sign at most four times on $\{1, \ldots, N-1\}$. The same holds for the second difference of f_N, and for f_N it holds for $N = 2$ as well.

We prove this in Appendix E.3.

The next theorem is the step that turns one sign into a whole convex stretch. Its idea is short. A sign change from negative to positive at $a_0 < a_1$, together with its mirror image, gives the signs $-,+,+,-$ at $a_0 < a_1 < N - a_1 < N - a_0$. A negative second difference in between would add a fifth alternating sign, which Theorem 3.16 forbids. A zero in between is subtler. We follow it back in time through (2), whose coefficients are positive, so a zero has two zero parents unless its parents have opposite signs, and opposite signs would again give five alternating signs. Going back to $t = 5$ forces $s_3 = s_4 = 0$, which Lemma 3.15 rules out.

Theorem 3.17 (no holes)**.** *Let $N \geq 5$ and $0 < \varepsilon < \frac{1}{2}$. Let $1 \leq a_0 < a_1 \leq N/2$ be integers with $D_N(a_0) < 0 < D_N(a_1)$. Then $D_N(a) > 0$ for every $a_1 \leq a \leq N - a_1$.*

We prove this in Appendix E.4.

Lemma 3.18 (the center from no holes)**.** *Let $N \geq 6$ be even and $0 < \varepsilon < \frac{1}{2}$. If $P_N(1) > \max\{E_N, C_N\}$, then $P_N(a) > C_N$ for $1 \leq a \leq N-1$, $a \neq h$, and $D_N(h) > 0$.*

Proof. Suppose that $P_N(a) \leq C_N$ for some $1 \leq a < h$, and let k be the first minimizer of P_N on $\{1, \ldots, h\}$. Then $2 \leq k < h$ and $P_N(k-1) > P_N(k) \leq P_N(k+1)$, so $D_N(k) > 0$. The first maximizer a_0 on $\{0, \ldots, k\}$ has value at least $P_N(1)$, which exceeds both endpoint values of this interval. Thus $1 \leq a_0 < k$ and $D_N(a_0) < 0$. Theorem 3.17 gives $D_N(a) > 0$ for $k \leq a \leq N - k$.

Put $\delta(a) = P_N(a+1) - P_N(a)$. These first differences are strictly increasing on $\{k-1, \ldots, N-k\}$, and symmetry gives $\delta(h) = -\delta(h-1)$. Hence $\delta(h-1) < 0$, and every difference from k to $h-1$ is negative. This makes $P_N(k) > P_N(h)$, a contradiction. Reflection proves the other half, and $D_N(h) = 2(P_N(h-1) - P_N(h)) > 0$. □

Corollary 3.19 (the shape at every even crossing)**.** *For every even $N \geq 8$, at the crossing,*

$$P_N(a) > C_N = E_N \quad (1 \leq a \leq N-1,\ a \neq h), \qquad D_N(h) > 0.$$

There are integers $1 \leq l \leq r < h$ such that P_N strictly increases from 0 to l, is constant from l to r, and strictly decreases from r to h. Its two intervals of global maximizers are $\{l, \ldots, r\}$ and $\{N - r, \ldots, N - l\}$.

Proof. At the crossing $P_N(1) > E_N = C_N$ by Proposition 3.7, so Lemma 3.18 gives the minimum and central curvature. For the humps, let $p = \mathrm{sp}(D_N(1), \ldots, D_N(h))$. It ends in $+$, and symmetry makes the full pattern p followed by its reverse, with the central positive blocks merged. If p has j blocks, the full pattern has $2j - 1 \leq 5$ by Theorem 3.16. Thus p is a subsequence of $(+,-,+)$.

The first differences $\delta(a) = P_N(a+1) - P_N(a)$ on $0 \le a < h$ therefore have at most three successive monotonic stretches, namely nondecreasing, nonincreasing, nondecreasing. Their first value is positive, and their last is $-D_N(h)/2 < 0$. The first stretch stays positive and the last stays negative. The middle stretch passes through zero at most once, possibly with consecutive zeros. This gives the claimed rise, plateau and fall. Symmetry gives the other hump. □

Corollary 3.20 (a window from the maximum)**.** *Let $N \ge 8$ be even and $x = x_N$. Let $[l, r]$ be the left interval of maximizers in Corollary 3.19, and put*

$$M = P_N(r), \qquad B = r - 1 + M^{-1}, \qquad b = \min\{h, \lceil B \rceil - 1\}.$$

Then $D_N(a) > 0$ for every integer $b \le a \le N - b$. This window is nonempty.

Proof. Let q be the first integer after l with $D_N(q) > 0$. Corollary 3.19 gives $D_N(l) < 0 < D_N(h)$, so $q \le h$. The constant plateau gives $q > r$. By its definition P_N is concave on $[r, q]$. Hence

$$\frac{M(q - r + 1)}{2} \le \sum_{a=r}^{q-1} P_N(a) \le \frac{1 - P_N(h)}{2} < \frac{1}{2},$$

where the first inequality sums the chord from $P_N(r) = M$ to the nonnegative value $P_N(q)$. Thus $q < B$, and consequently $q \le b \le h$. Theorem 3.17, applied at $l < q$, gives positivity on $[q, N - q]$, which contains $[b, N - b]$. □

The finite endpoint-dip certificates in Proposition 3.7 remain inputs. Neither corollary needs a mode-location or convex-window estimate. The height bound gives a window at every even length, although it may consist only of the center. Heuristically the window is much wider. Near the crossing, Theorem 3.9(b) and Corollary 3.11 give $M = P_N(r) = \frac{f_L(\lambda_L)}{2x}(1 + o(1))$, so with $m^* = r$ we get $B = m^* + \big(2/f_L(\lambda_L) + o(1)\big)x$. Since $f_L(\lambda_L) \approx 0.1807$, this is about $m^* + 11x$, far below h for large N. At $N = 4000$ the computed $B - m^*$ is about $9.9x$. We do not compute the $o(1)$, and the constants of Theorem 3.9 are far from sharp, so this is not a bound. A stronger, prescribed window is studied in the arXiv version of this paper [41].

3.6 Odd N

Proposition 3.21 (odd N)**.** *Let $N \ge 3$ be odd, and write $\widetilde{C}_N = P_N(\frac{N-1}{2}) = P_N(\frac{N+1}{2})$ for the two central values. The ratio $\widetilde{C}_N / E_N$ is strictly increasing in ε, and it equals 1 at exactly one $\varepsilon_N \in (0, \frac{1}{2})$. Put $x_N = N\varepsilon_N$.*

(a) For every $0 < \varepsilon < \frac{1}{2}$,

$$\widetilde{C}_N = \Big(1 - \frac{c}{N}\Big)^{-1} C_{N+1},$$

where C_{N+1} is the center of the walk with $N + 1$ steps and the same ε.

The center of a walk with an even number of steps is reached from the two central values one step earlier, and from each of them with the same probability. So the two central values of an odd N are the center of $N + 1$ up to the factor $1 - c/N$, and what we know about the crossing for even N carries over. We show in the arXiv version of this paper [41] that the crossing for odd N satisfies Theorem 2.7, that $P_N(0) < P_N(1)$ at the crossing for odd $N \ge 1689$, and that the convexity window of the arXiv version of this paper [41] holds at the crossing for every large odd N.

Proof. The crossing. By Proposition 2.3, $P_N(k)/E_N$ is strictly increasing in ε for $k = \frac{N+1}{2}$. It tends to 0 as $\varepsilon \to 0$. At $\varepsilon = \frac{1}{2}$ the steps are independent and fair, so $\widetilde{C}_N = \binom{N}{k} 2^{-N} > 2^{-N} = E_N$. So the ratio equals 1 at exactly one ε_N, and $\varepsilon_N < \frac{1}{2}$.

(a) Put $k = \frac{N+1}{2}$. Since $\varepsilon + \frac{c}{2} = \frac{1}{2}$, we have $r_N(k) = \frac{1}{2} + \frac{c}{2N}$ and $r_N(k-1) = \frac{1}{2} - \frac{c}{2N}$. So (1) with $t = N$ and $a = k$ gives

$$C_{N+1} = P_{N+1}(k) = \Big(\frac{1}{2} - \frac{c}{2N}\Big)\big(P_N(k) + P_N(k-1)\big) = \Big(1 - \frac{c}{N}\Big)\widetilde{C}_N.$$

□

4 The fixed-memory limit law

In this section the reversal probability $\varepsilon \in (0, \frac{1}{4})$ is fixed. So the memory parameter $p = 1 - \varepsilon$ is fixed too, and $a = 2p - 1 = 1 - 2\varepsilon$ lies in $(\frac{1}{2}, 1)$. We write $S_k = \sum_{j \le k} X_j$, so that $S_N = 2A_N - N$. If $X_1 = 1$, then S_k/k^a converges almost surely to a random variable $\mathcal{L}_a$ [5, Theorem 3.7]. With a fair first step the walk given $X_1 = -1$ is the mirror image of the walk given $X_1 = 1$. So S_k/k^a converges almost surely to a random variable $\mathcal{L}_a^\circ$ with the law of $\xi\mathcal{L}_a$, where ξ is a fair sign independent of $\mathcal{L}_a$ [8, (1.3) and (1.6)]. Guérin, Laulin and Raschel prove that both limits have positive, bounded and smooth densities [8, Theorem 1.3]. We write g_a for the density of $\mathcal{L}_a^\circ$, which is even. Guérin, Laulin, Raschel and Simon prove that $\mathcal{L}_a$ is unimodal [7, Theorem 1.1].

4.1 Memory near one

Corollary 4.1 (the fixed-memory limit near one)**.** *For the elephant walk with $X_1 = 1$ and fixed memory $p = (1 + a)/2$, $a \in (\frac{1}{2}, 1)$, let $\mathcal{L}_a = \lim_{k\to\infty} S_k/k^a$ as in [5, Theorem 3.7]. If Z has characteristic function $\mathbb{E}e^{itZ} = \exp\{-it \log|t| - \pi|t|/2\}$, continuously extended at $t = 0$, then*

$$\frac{1 - \mathcal{L}_{1-2\varepsilon}}{2\varepsilon} + \log(1/\varepsilon) \implies Z \qquad (\varepsilon \downarrow 0).$$

Proof. Fix $0 < \varepsilon < 1/8$, put $a = 1 - 2\varepsilon$, and use one infinite walk with $X_1 = 1$. By [5, (2.4), Appendix A], $\mathcal{M}_k = \Gamma(k)S_k/\Gamma(k + a)$ is a martingale with limit $\mathcal{L}_a$. It is Bercu's martingale divided by $\Gamma(1 + a)$. Its increments are

$$\mathcal{M}_{k+1} - \mathcal{M}_k = \frac{\Gamma(k+1)}{\Gamma(k+1+a)}\left(X_{k+1} - \frac{aS_k}{k}\right).$$

Log-convexity of gamma gives $1 \le k^a\Gamma(k)/\Gamma(k+a) \le (1+a/k)^{1-a} \le 1+1/k$. Since $S_k = k - 2M_k$, Lemma B.2 gives

$$\mathbb{E}\big[1 - a^2(S_k/k)^2\big] \le 4\varepsilon + \frac{4\mathbb{E}M_k}{k} \le 4\varepsilon(1 + \log k).$$

Orthogonality of the increments therefore gives square-mean convergence and

$$\mathbb{E}|\mathcal{L}_a - \mathcal{M}_n|^2 \le C\varepsilon \sum_{k \ge n} k^{-2a}(1 + \log k) \le C\varepsilon n^{1-2a}(1 + \log n),$$

with C independent of $a \in [3/4, 1)$, since $2a - 1 \ge 1/2$.

Take $n = \lfloor \varepsilon^{-4/3} \rfloor$ and $x = n\varepsilon$. Then $x \to \infty$ and $x^3 \log^2 n/n = n^2\varepsilon^3 \log^2 n \to 0$, so Theorem 3.9(c) gives $Z_\varepsilon = (M_n - x\log x)/x \Rightarrow Z$. The tail bound gives $\mathbb{E}|(\mathcal{L}_a - \mathcal{M}_n)/\varepsilon|^2 = O(\varepsilon^{1/3}(1 + \log(1/\varepsilon))) = o(1)$, using $n^{4\varepsilon} \to 1$. Also $|\mathcal{M}_n - n^{-a}S_n| \le n^{-a} = o(\varepsilon)$. Finally,

$$n^{-a}S_n = e^{2\varepsilon \log n}[1 - 2\varepsilon(\log x + Z_\varepsilon)],$$
$$\frac{1 - n^{-a}S_n}{2\varepsilon} + \log(1/\varepsilon) = Z_\varepsilon + o_{\mathbb{P}}(1).$$

The second equality uses $\log x = \log n + \log \varepsilon$, tightness of Z_ε, and $\varepsilon \log^2 n \to 0$. Replacing $n^{-a}S_n$ by $\mathcal{L}_a$ proves the claim. □

This is weak convergence. Trinas and Valle find $\operatorname{Var} \mathcal{L}_a = 2(1-a) + o(1-a)$ [9, Eq. (4.5)]. Both results say that $\mathcal{L}_a$ concentrates at 1 as $a \to 1$, and we use this below.

4.2 The number of modes

A probability density h on $\mathbb{R}$ is *unimodal* if for some m it is nondecreasing on $(-\infty, m]$ and nonincreasing on $[m, \infty)$. Guérin, Laulin, Raschel and Simon ask for a unique local maximum [7], which also rules out a flat top. Our g_a mixes the unimodal density of $\mathcal{L}_a$ with its mirror image, and unimodality is not preserved by mixtures [9, Section 4]. Guérin, Laulin, Raschel and Simon write L_q for the limit of S_n/n^a when $\mathbb{P}(X_1 = 1) = q$, and state the following conjecture [7, Remark 2.1(b)].

> Simulations show that in the case $q \in (0,1)$, the random variable L_q is not unimodal in general. By (1) and Theorem 1.1 the density of L_q is the superposition of two unimodal functions and we conjecture that for every $a \in (1/2, 1)$ there exists $q(a) \in [0, 1/2]$ such that L_q is bimodal if $|q - 1/2| < q(a)$ and unimodal if $|q - 1/2| \ge q(a)$. This problem is believed to be challenging.

Here (1) is the identity $L_q = (2\xi_q - 1)L_1$ in law, with ξ_q Bernoulli with parameter q, and $L_{1/2} = \mathcal{L}_a^\circ$. At $q = \frac{1}{2}$ the condition $|q - \frac{1}{2}| < q(a)$ says $q(a) > 0$. So for a fair first step the conjecture says that $\mathcal{L}_a^\circ$ is unimodal or bimodal. The next theorem proves this for every a, with modes that may be intervals.

Theorem 4.2 (at most two modes)**.** *Let $a \in (\frac{1}{2}, 1)$. Exactly one of the following holds.*

(a) The density g_a is nondecreasing on $(-\infty, 0]$ and nonincreasing on $[0, \infty)$.

(b) There is $c > 0$ such that g_a is nondecreasing on $[0, c]$ and nonincreasing on $[c, \infty)$, and $g_a(0) < g_a(c)$.

In case (b), g_a is nondecreasing on $(-\infty, -c]$ and nonincreasing on $[-c, 0]$, and it is not unimodal. The only possible strict local maximum of g_a is 0 in case (a), and the only possible ones are $\pm c$ in case (b). If g_a is constant on no interval, the same holds for every local maximum.

So g_a has at most two modes, and in case (b) we say that it has exactly two. In the proof a Gaussian kernel turns the second differences of P_N into a smooth function with at most their sign pattern, by Descartes' rule of signs for exponential sums. Then $N \to \infty$, and then the width of the kernel tends to 0. The lemmas are proved in Appendix F. A real function v has *values at most* Π if $\operatorname{sp}(v(y_1), \dots, v(y_j)) \preceq \Pi$ for all $y_1 < \cdots < y_j$, and *increments at most* Π if $\operatorname{sp}(v(y_2) - v(y_1), \dots, v(y_j) - v(y_{j-1})) \preceq \Pi$ for all such points. We put $\Pi_4 = (+, -, +, -)$.

Proof. Fix a, so that $\varepsilon = (1-a)/2 \in (0, \frac{1}{4})$.

Step 1, smoothing. Fix $\sigma > 0$ and let $\varphi(y) = e^{-y^2/(2\sigma^2)}/(\sigma\sqrt{2\pi})$. For $N \geq 3$ put $\eta = 2N^{-a}$ and $z_k = (2k-N)N^{-a}$, so that $S_N/N^a = z_{A_N}$, and let

$$V_N(y) = \eta^{-2} \sum_{k=1}^{N-1} D_N(k)\, \varphi(y - z_k).$$

Since $\varphi(y - z_k) = \varphi(y) e^{-z_k^2/(2\sigma^2)} e^{y z_k/\sigma^2}$ and the z_k increase with k, Lemma F.1 and Theorem 3.16 show that V_N has values at most Π_5. (If every $D_N(k)$ vanishes, then $V_N = 0$ and every pattern is empty.)

Step 2, the limit $N \to \infty$. Let $u(y) = \mathbb{E}\varphi(y - \mathcal{L}_a^\circ)$. The derivatives of φ are bounded, so u is smooth and $u^{(j)}(y) = \mathbb{E}\varphi^{(j)}(y - \mathcal{L}_a^\circ)$. We show that $V_N(y) \to u''(y)$ for every y. Put $P_N(k) = 0$ for $k \notin \{0, \dots, N\}$. Then $D_N(k) = 0$ unless $-1 \leq k \leq N+1$, and summation by parts gives

$$\sum_{k=-1}^{N+1} D_N(k)\, \varphi(y - z_k) = \sum_{k=0}^{N} P_N(k)\, \Delta\varphi(y - z_k), \qquad \Delta\varphi(t) = \varphi(t+\eta) - 2\varphi(t) + \varphi(t-\eta).$$

Taylor's formula gives $|\eta^{-2}\Delta\varphi(t) - \varphi''(t)| \leq \eta^2 \|\varphi''''\|_\infty/12$. By symmetry the four terms with $k \in \{-1, 0, N, N+1\}$ on the left are at most $\varphi(0)\,(6P_N(0) + 2P_N(1))$ in absolute value. Here $P_N(0) \leq (1-\varepsilon)^{N-1}$, and the proof of Theorem 3.5 gives $f_N(1) \leq \varepsilon N(1-\varepsilon)^{N-2}$ and $f_N(N-1) \leq \varepsilon(1-\varepsilon)^{N-2}$, so that $P_N(1) \leq \varepsilon N(1-\varepsilon)^{N-2}$. As $\eta^{-2} = N^{2a}/4$, these four terms times η^{-2} tend to 0. Hence

$$V_N(y) = \mathbb{E}\varphi''\big(y - S_N/N^a\big) + o(1) \longrightarrow \mathbb{E}\varphi''(y - \mathcal{L}_a^\circ) = u''(y),$$

because $S_N/N^a \to \mathcal{L}_a^\circ$ in law and φ'' is bounded and continuous. By Lemma F.2, u'' has values at most Π_5.

Step 3, one derivative fewer. By dominated convergence, $u'(y) \to 0$ as $y \to \pm\infty$. So u has increments at most Π_4, by Lemma F.3.

Step 4, the limit $\sigma \to 0$. Write $u = u_\sigma$. As g_a is bounded and continuous, $u_\sigma(y) = \int g_a(y - t)\varphi(t)\, dt$ tends to $g_a(y)$ for every y. So the increments of u_σ tend to those of g_a, and Lemma F.2 shows that g_a has increments at most Π_4.

Step 5, symmetry. Let $0 \leq y_1 < \cdots < y_j$, let $d_i = g_a(y_{i+1}) - g_a(y_i)$, and let p be the pattern of the d_i. As g_a is even, the points $-y_j < \cdots < -y_1 < y_1 < \cdots < y_j$, with $-y_1$ left out if $y_1 = 0$, have the increments $-d_{j-1}, \dots, -d_1$, then 0 if $y_1 > 0$, then $d_1, \dots, d_{j-1}$. Their pattern is the reverse of $-p$ followed by p. It has $2|p|$ entries and is $\preceq \Pi_4$. So p is empty, $(+)$, $(-)$ or $(+,-)$. In particular there are no $0 \leq y < y' < y''$ with $g_a(y) > g_a(y') < g_a(y'')$.

If no increment on $[0, \infty)$ is positive, (a) holds by evenness. Otherwise $g_a(y_1) < g_a(y_2)$ for some $0 \leq y_1 < y_2$. Some increment on $[y_2, \infty)$ is negative, since otherwise $g_a \geq g_a(y_2) > 0$ there, against $\int g_a = 1$. If $g_a(y_4) < g_a(y_3)$ with $y_2 \leq y_3 < y_4$, then g_a cannot rise after y_4, so it attains its maximum over $[0, \infty)$ at some $c \in [0, y_4]$. If $0 \leq y < y' \leq c$ and $g_a(y) > g_a(y')$, then $g_a(y') < g_a(c)$, and (y, y', c) is a fall followed by a rise. If $c \leq y < y'$ and $g_a(y) < g_a(y')$, then (c, y, y') is one. So g_a is nondecreasing on $[0, c]$ and nonincreasing on $[c, \infty)$. If $g_a(0) = g_a(c)$, then g_a would be nonincreasing on $[0, \infty)$, against $g_a(y_1) < g_a(y_2)$. So (b) holds.

As $g_a(0) < g_a(c)$ in (b), the two cases exclude each other. In (b), symmetry gives the monotonicity on $(-\infty, 0]$. If g_a were unimodal with mode m, then $m \geq 0$ would give $g_a(-c) \leq g_a(0)$ and $m < 0$ would give $g_a(0) \geq g_a(c)$, while $g_a(\pm c) > g_a(0)$. Finally a local maximum y other than the points named lies in an interval on which g_a is monotone, so g_a is constant on a one-sided neighborhood of y, and y is not strict. □

Theorem 4.3 (two modes for memory near one)**.** *If* $\frac{39}{40} \le a < 1$*, that is* $0 < \varepsilon \le \frac{1}{80}$*, then case (b) of Theorem 4.2 holds. So* g_a *has exactly two modes.*

Proof. Put $\delta = 1 - a \le \frac{1}{40}$, $\mu = \mathbb{E}\mathcal{L}_a$ and $v = \operatorname{Var}\mathcal{L}_a$. The moment recursion of [8, Theorem 1.4], with $k = 1$ and $k = 2$, gives $\mu = 1/\Gamma(1+a)$ and $\mathbb{E}\mathcal{L}_a^2 = 1/((2a-1)\Gamma(2a))$, as also recorded in [9, Eq. (4.3)]. The function $\log\Gamma$ is convex on $(0,\infty)$, since $\psi' > 0$ there [40, 5.15.1]. Also $\Gamma(1) = \Gamma(2) = 1$ and $\psi(2) = 1-\gamma$ [40, 5.4.1, 5.4.14]. So $0 \ge \log\Gamma(y) \ge (1-\gamma)(y-2)$ for $1 \le y \le 2$. With $y = 1+a$ and $y = 2a$ this gives

$$1 \le \mu \le M := e^{(1-\gamma)\delta}, \qquad v \le \mathbb{E}\mathcal{L}_a^2 - 1 \le V := \frac{e^{2(1-\gamma)\delta}}{1-2\delta} - 1.$$

Both M and V increase with δ, so $M \le 1.0107$ and $V \le 0.0752$.

Suppose that (a) holds. Then $F(t) = \mathbb{P}(|\mathcal{L}_a^\circ| \le t) = 2\int_0^t g_a$ is concave on $[0,\infty)$ with $F(0) = 0$, so $\frac{5}{2}F(\frac{2}{5}) \ge \frac{4}{7}F(\frac{7}{4})$. Also $|\mathcal{L}_a^\circ|$ has the law of $|\mathcal{L}_a|$. Cantelli's inequality $\mathbb{P}(\mathcal{L}_a \le \mu - w) \le v/(v+w^2)$, which is Chebyshev's inequality for $\mu - \mathcal{L}_a + v/w$, with $w = \mu - \frac{2}{5} \ge \frac{3}{5}$ gives

$$F(\tfrac{2}{5}) \le \mathbb{P}(\mathcal{L}_a \le \tfrac{2}{5}) \le \frac{V}{V + 9/25} \le 0.173.$$

If $|\mathcal{L}_a| > \frac{7}{4}$, then $|\mathcal{L}_a - \mu| > \frac{7}{4} - \mu \ge \frac{7}{4} - M$, so Chebyshev's inequality gives $F(\frac{7}{4}) \ge 1 - V/(\frac{7}{4}-M)^2 \ge 0.862$. Hence $\frac{5}{2}F(\frac{2}{5}) \le 0.433 < 0.492 \le \frac{4}{7}F(\frac{7}{4})$, a contradiction. So (b) holds. □

Theorem 4.2 proves the case $q = \frac{1}{2}$ of the conjecture of Guérin, Laulin, Raschel and Simon for every a, up to flat pieces. A mode may be an interval on which g_a is constant, and we do not rule this out. Theorem 4.3 shows that the bimodal case occurs, and its proof uses no computer. The next theorem locates the change from case (a) to case (b). Its proof is computer-assisted, and it covers the rest of $(\frac{1}{2}, 1)$.

Theorem 4.4 (the threshold, computer-assisted)**.** *Put*

$$z_- = 1838418790980431/2748779069440000 = 0.66881286001459775\ldots,$$
$$z_+ = 45960469774511/68719476736000 = 0.66881286001460103\ldots,$$

so that $z_+ - z_- < 3.3 \cdot 10^{-15}$.

(a) For every $a \in (\frac{1}{2}, z_-]$ *we have* $g_a''(0) < 0$*, and case (a) of Theorem 4.2 holds. So* g_a *is unimodal.*

(b) For every $a \in [z_+, 1)$ *case (b) holds, so* g_a *has exactly two modes. For* $a \in [z_+, 0.6712]$ *also* $g_a''(0) > 0$.

(c) The function $a \mapsto g_a''(0)$ *has a zero in* (z_-, z_+)*, and every zero in* $(\frac{1}{2}, 0.6712]$ *lies there.*

So the supremum of the a with case (a) and the infimum of the a with case (b) both lie in $[z_-, z_+]$. This settles the case $q = \frac{1}{2}$ of the conjecture, up to flat modal intervals, with the threshold located to within $3.3 \cdot 10^{-15}$. If the conjecture holds, then $q(a) = 0$ for $a \le z_-$ and $q(a) > 0$ for $a \ge z_+$. We do not prove that $g_a''(0)$ has only one zero in (z_-, z_+), and we do not know which case holds there. Theorem 4.3 gives part (b) for $a \ge \frac{39}{40}$ without a computer, and the certificate covers the rest. The conjecture for $q \ne \frac{1}{2}$ remains open. For $q \ne \frac{1}{2}$ the finite law $qf_N(N-k) + (1-q)f_N(k)$ is not symmetric, and we have no analogue of Theorem 3.16 for it.

Remark 4.5 (the method). Appendix G gives the proof, and we sketch it here. The moments of $\mathcal{L}_a$ follow from the exact recursion of [8, Theorem 1.4], and $\mathcal{L}_a$ has a finite moment generating function [8, Corollary 2.11]. So its characteristic function ϕ_a is the sum of its moment series, and $g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\phi_a(t)\,dt$. The fixed-point equation of [8, Theorem 1.1] gives an inequality for $|\phi_a|$ that improves itself. From bounds on a finite range of t it yields $|\phi_a(t)| \le Ct^{-4}$ for all larger t, with an explicit C that holds for a whole interval of a. The moment series cancels terms of size about 10^{60}, so we cannot sum it with a as an interval. Instead we expand in a around the centre of each interval, in ball arithmetic at that exact point, and bound the remainder by Cauchy's estimate on a complex disc. This is a Taylor model in a. Near $a = \frac{1}{2}$ we use the scaled variable $\mathcal{L}_a/(\mathbb{E}\mathcal{L}_a^2)^{1/2}$, whose moments are analytic at $a = \frac{1}{2}$, so the cover reaches $\frac{1}{2}$. For $a \ge 0.6705$ we rule out case (a) by moments alone. In case (a), Khinchine's theorem would make $(2j+1)\mathbb{E}\mathcal{L}_a^{2j}$ a moment sequence of a positive variable, and a Hankel quadratic form of order at most 56 is negative instead. The certificate takes about 2.7 hours of CPU time. On the exact finite law, $D_N(N/2)$ changes sign at $a \approx 0.6563$ for $N = 4000$ and at $a \approx 0.6592$ for $N = 8000$. These two numbers are computed and are not used in any proof.

5 Verification

The proofs use four certified computations, apart from short exact ratios such as those in Proposition 3.7. First, `certify_elephant.py` encloses the 60-term sum in Theorem 2.6(b) in $[102.586396, 102.586398]$. With the analytic tail, it is below 102.59, giving the constant 5200. Any finite bound suffices asymptotically, but this value determines the explicit thresholds in Proposition 3.7 and the second-order results. Second, the same script certifies the Landau derivative and curvature enclosures of Lemma 3.10. Lean checks these independently.

Third, `certify_center.py` supplies the finite endpoint-dip input of Proposition 3.7 for $10 \le N \le 2000$. It brackets each crossing and checks by an exact rational comparison that the lower end of the bracket exceeds $2/(N+2)$. It also checks the center minimum independently, which Corollary 3.19 no longer needs. The script `certify_convex.py` supports only the finite convexity window of the arXiv version (the arXiv version of this paper [41]). The outputs and rational witnesses are in `data/certify_center_output_2026-10-17.txt`, `data/certify_convex_output_2026-10-17.txt`, `data/witnesses_center_2026-10-17.jsonl` and `data/witnesses_convex_2026-10-17.jsonl`. The witnesses record script hashes and software versions. Each certification script fails if a required check fails, and range-based scripts also reject empty ranges.

Fourth, `certify_threshold.py` proves Theorem 4.4 as described in Appendix G. It imports `threshold_taylor.py`, `threshold_hankel.py` and `threshold_pointwise.py`. Its 9 runs are recorded in `data/threshold_*.txt`, each with a header that records the hashes of the 4 scripts and the software versions, and with the exit status as its last line. Its mode `check` verifies that the records join into the cover used in the proof. Theorems 4.2 and 4.3 use no computation. The script `verify_elephant_modes.py` checks the constants in the proof of Theorem 4.3 and computes the finite-N numbers of Remark 4.5, which are not proof inputs. Its output, with a header that records the hash of the script and the software versions, is in `data/verify_modes_output.txt`.

The Lean 4.33.1 library in `kagey-problems/problem131/lean/PaperD` proves the finite identities, crossing bounds, sign arguments and certificate checks. The existing formal bulk and eventual-shape proofs use six named classical inputs, namely Fourier inversion, the compound Poisson characteristic function, Efron–Stein, Scheffé, unimodality of class-L laws, and Hoeffding's hypergeometric bound. Their sources and precise formal status are in `MAP.md` and `coverage_tables.tex`. The library uses Mathlib revision `0df444a360ea`. The recorded kernel build reports only `propext`,

`Classical.choice` and `Quot.sound`. Generated certificate entries are checked with exact outward-rounded integer arithmetic, without `native_decide`. This does not formalize the six classical inputs. Lemma 3.18 and the full all-even shape in Corollary 3.19 are kernel-checked, including both complete intervals of maximizers. The composed theorem `shape_at_crossing_all_even` needs only $N = 2h$ with $h \geq 4$. The maximum-height window in Corollary 3.20, the total variation strengthening of Theorem 3.8, the fixed-memory bridge and the results of Section 4.2 are also not yet formalized. The library's earlier pointwise compound Poisson bound remains valid.

The separate Python runs used Python 3.14.3, python-flint 0.9.0 (FLINT 3.6.0, including Arb), mpmath 1.3.0, numpy 2.4.2 and scipy 1.17.1. The figures, tables and other numerical illustrations are not proof inputs. Further checks and rerun commands are documented in the repository named under Code availability.

Acknowledgments. I thank Peter Kagey, whose Problem 131 of the Open Problem Collection [10] asks when the center and the ends of the persistent walk are equally likely, and Aidan Deshong for insightful conversations.

A note on the use of AI

I wrote the original draft and its mathematics. I used Claude Opus 5.5 (Anthropic) for revision and proof checking, and later used OpenAI Codex to revise the exposition and check references. This included the abstract, introduction, description of the verification and the arrangement of the appendices and of the further results in the arXiv version. Both models were used to expand literature search beyond what I had included, and all sources were read by me before citation if appropriate. AI tools also generated the Lean files and parts of the verification and certificate code, which I reviewed against the paper. Section 5 distinguishes the formal proofs, their named classical hypotheses and the separate numerical checks. I take responsibility for all mathematical claims, proofs and computations.

Statements and Declarations

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Data availability. The outputs of the computations described in Section 5 are in the folder `kagey-problems/problem131/paperD/data` of the repository named under Code availability. The arXiv version of this paper [41] contains further results, among them the center and the crossing to second order, the tail and a convexity window between the humps. This article does not use them. That version also gives the full proofs of the steps in Appendix G. There is no supplementary material.

Code availability. All code, data and Lean files supporting this paper are in the folder `kagey-problems/problem131` of https://github.com/apemm/Kagey-Problems. The supporting code, certificate records and Lean files are those at commit `aa88996f79dc` of that repository, including the certificate for Theorem 4.4. To rerun the checks, run `python -B verify_all.py --paper d` in that folder, which takes several minutes. The certified computation behind Proposition 3.7 takes about half an hour and runs separately in `paperD` as `python -B certify_center.py 10 2000`. The certified computation behind the finite convexity window of the arXiv version takes about an hour and runs as `python -B certify_convex.py 10 2000`. The certificate behind Theorem 4.4 takes about 2.7 hours of CPU time, or 29 minutes in 9 parallel processes, and runs as `python -B certify_threshold.py full`. The scripts need `numpy`, `scipy`, `mpmath` and `python-flint`, and the file `SUPPLEMENT.md` says what each script checks.

Use of AI tools. See the note on the use of AI tools just before these declarations.

Author contributions. I am the sole author and am responsible for the whole paper.

A Proofs for Section 2

We prove Theorem 2.6. Throughout, $N \geq 2$ is even and we condition on $X_1 = +1$. By Proposition 2.5(a) we need to bound $V_{t+1}(1)$, and we compare V with explicit hitting probabilities of Pólya urns. Let $\mathcal{L}_s F(b) = (1-r_s(b))F(b)+r_s(b)F(b+1)$, so that $V_s = \mathcal{L}_s V_{s+1}$ and $V_N(b) = \mathbf{1}\{b = h\}$. Both M_t and $t - M_t \geq 1$ are nondecreasing. So from $b \geq 1$ the chain stays at states ≥ 1, and for $1 \leq b \leq s-1$ we have $V_s(b) = 0$ unless b is feasible, that is, $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \leq b \leq h,\ 1 \leq s - b \leq h\}$. Note that $\mathcal{S}_N = \{h\}$.

For $c \geq 0$ the Pólya(c) chain moves at time s from b to $b+1$ with probability $r_s^c(b) = \frac{b+c}{s+2c}$. It is Pólya's urn with weights $b+c$ and $s-b+c$ for the two colors, that is, with c extra balls of each color. For $b+c > 0$ and $s-b+c > 0$ let $\Phi_s^c(b) = \binom{N-s}{h-b}\mathrm{B}(h+c, h+c)/\mathrm{B}(b+c, s-b+c)$, where B is the beta function and the binomial coefficient is 0 unless $0 \leq h-b \leq N-s$. Theorem 2.6(a) will follow from Lemma A.1(d), and (b) from Proposition A.4.

Lemma A.1. *Let $c \geq 0$, $1 \leq s \leq N$, $b+c > 0$ and $s-b+c > 0$.*

(a) The number $\Phi_s^c(b)$ is the probability that the Pólya(c) chain at b at time s is at h at time N. So $\Phi_N^c(b) = \mathbf{1}\{b = h\}$, and $\Phi_s^c(b) = r_s^c(b)\Phi_{s+1}^c(b+1) + (1 - r_s^c(b))\Phi_{s+1}^c(b)$ for $s < N$.

(b) If $s < N$ and $\Phi_s^c(b) > 0$, then $r_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \bar{r}_s(b) := \frac{h-b}{N-s}$.

(c) If $0 < \varepsilon < \frac{1}{2}$ and $\alpha = \frac{\varepsilon}{1-2\varepsilon}$, then $r_s(b) = r_s^{\alpha s}(b)$ for $0 \leq b \leq s$.

(d) If $0 < \varepsilon < \frac{1}{2}$ and $0 \leq c \leq \alpha s$, then $V_s(b) \geq \Phi_s^c(b)$ for $1 \leq b \leq s-1$. In particular $V_{t+1}(1) \geq \Phi_{t+1}^0(1) = \pi_t$ for $1 \leq t \leq h$.

Proof. (a) Let $x = b+c$, $y = s-b+c$ and $(x)_k = x(x+1)\cdots(x+k-1)$. As in the proof of Lemma 2.4, each order of the $N-s$ draws with $h-b$ draws of the first color has probability $(x)_{h-b}(y)_{N-s-h+b}/(x+y)_{N-s} = \mathrm{B}(h+c, h+c)/\mathrm{B}(x, y)$, and there are $\binom{N-s}{h-b}$ such orders. The recursion is the Markov property.

(b) Use $\binom{N-s-1}{h-b-1}/\binom{N-s}{h-b} = \frac{h-b}{N-s}$ and $\mathrm{B}(x, y)/\mathrm{B}(x+1, y) = \frac{x+y}{x} = 1/r_s^c(b)$.

(c) Since $1 + 2\alpha = 1/(1-2\varepsilon)$, we have $\frac{b+\alpha s}{s+2\alpha s} = (1-2\varepsilon)(\frac{b}{s} + \alpha) = r_s(b)$.

(d) We check the hypotheses of Lemma A.2(b) below for $F_u = \Phi_u^c$ on $\{1, \dots, u-1\}$, where $s \le u \le N$. The reason is that M has $\alpha u \ge c$ extra balls at time u, so it is pulled toward $\frac{u}{2}$ at least as strongly as the Pólya(c) chain, and from a state closer to $\frac{u}{2}$ it is easier to end at h.

By (a), $F_N(b) = \mathbf{1}\{b = h\}$, and F_u vanishes off $\mathcal{S}_u$ because the binomial coefficient does. Let $s \le u < N$ and $b \in \mathcal{S}_u$. Subtracting the recursion in (a) from the definition of $\mathcal{L}_u$ gives

$$\mathcal{L}_u F_{u+1}(b) - F_u(b) = \big(r_u(b) - r_u^c(b)\big)\big(F_{u+1}(b+1) - F_{u+1}(b)\big).$$

We show that the two factors on the right never have opposite signs. For the first factor, $r_u^c(b) = \frac{1}{2} - (\frac{u}{2} - b)/(u + 2c)$, and by (c) the same formula with αu in place of c gives $r_u(b)$. So the first factor is $\frac{u}{2} - b$ times $\frac{1}{u+2c} - \frac{1}{u+2\alpha u}$. The second of these is nonnegative because $c \le \alpha s \le \alpha u$. Hence the first factor has the sign of $\frac{u}{2} - b$ or vanishes. For the second factor there are three cases. If $b = h$ then $b \ge \frac{u}{2}$ and $F_{u+1}(b+1) = 0$, so the factor is ≤ 0. If $u - b = h$ then $b \le \frac{u}{2}$ and $F_{u+1}(b) = 0$, so the factor is ≥ 0. Otherwise both values are positive, and the formula for Φ gives $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(h-b)(u-b+c)}{(h-u+b)(b+c)}$. With $b = \frac{u}{2} - d$ the numerator minus the denominator is $2d(h+c)$, so the ratio is ≥ 1 if $b \le \frac{u}{2}$ and ≤ 1 if $b \ge \frac{u}{2}$. In all three cases the second factor has the sign of $\frac{u}{2} - b$ or vanishes. Hence $\mathcal{L}_u F_{u+1} \ge F_u$ on $\mathcal{S}_u$, and Lemma A.2(b) with $s_0 = s$ gives $V_s \ge F_s$. Finally $\Phi_{t+1}^0(1) = \beta_{t+1}(h) = \pi_t$ by (a) and Lemma 2.4. □

Lemma A.2. *Let $1 \le s_0 \le N$, and for $s_0 \le s \le N$ let $F_s \ge 0$ be a function on $\{1, \dots, s-1\}$.*

(a) If $F_N(h) \ge 1$ and $\mathcal{L}_s F_{s+1} \le K_s F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, with constants $K_s \ge 0$, then $V_s \le \big(\prod_{u=s}^{N-1} K_u\big) F_s$ on $\mathcal{S}_s$.

(b) If $F_N(b) \le \mathbf{1}\{b = h\}$, each F_s vanishes off $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, then $V_s \ge F_s$.

Proof. We use backward induction on s, from $V_N(b) = \mathbf{1}\{b = h\}$ and $V_s = \mathcal{L}_s V_{s+1}$. Note that $\mathcal{L}_s$ is an average of two values, so it keeps inequalities between functions. In (a), suppose that $V_{s+1} \le \prod_{u>s} K_u \, F_{s+1}$ on $\mathcal{S}_{s+1}$. This also holds off $\mathcal{S}_{s+1}$, because there $V_{s+1} = 0$ and $F_{s+1} \ge 0$. So on $\mathcal{S}_s$ we get $V_s = \mathcal{L}_s V_{s+1} \le \prod_{u>s} K_u \, \mathcal{L}_s F_{s+1} \le \prod_{u \ge s} K_u \, F_s$. In (b), suppose that $V_{s+1} \ge F_{s+1}$ on $\{1, \dots, s\}$. Then $V_s = \mathcal{L}_s V_{s+1} \ge \mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$, and $F_s = 0 \le V_s$ off $\mathcal{S}_s$. □

Idea of the proof of the upper bound. We compare M with an urn whose number of extra balls, $\kappa(s+10)$ of each color at time s, grows more than twice as fast as the αs of M. Its hitting probability $\Phi_s^{\kappa_s}$ is explicit. One step of M applied to it gives it back times a factor Δ_s, and Proposition A.4 shows that $\log \Delta_s \le \kappa/s$ up to a smaller term. Two effects enter Δ_s. Raising the extra balls from κ_s to κ_{s+1} changes Φ, and we bound this through the derivative of Φ_s^τ in τ (Lemma A.3). Also M pulls toward the middle less than the comparison urn, which helps. The product of the factors is about $e^{\kappa H_N}$. Let ψ be the digamma function.

Lemma A.3. *Let $2 \le s \le N$, $b \in \mathcal{S}_s$ and $\tau \ge 0$. Put $X = b + \tau$, $Y = s - b + \tau$, $\Sigma = X + Y$ and $z = (s - 2b)^2/\Sigma^2$. Then*

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1 - z) + \frac{1 + z}{(1 - z)\Sigma} + \frac{2(1 + z)}{3\Sigma^2(1 - z)^2}.$$

Proof. We differentiate the beta functions and bound the digamma function by the first terms of its expansion. Only the two beta functions in $\Phi_s^\tau(b)$ depend on τ, and $\log \mathrm{B}(x, y) = \log \Gamma(x) + \log \Gamma(y) - \log \Gamma(x + y)$. So the derivative is $2\psi(\Sigma) - \psi(X) - \psi(Y)$ minus the difference $2\psi(N + 2\tau) - 2\psi(h + \tau)$.

The duplication formula [40, 5.5.5] gives $2\psi(2y) = \psi(y) + \psi(y+\frac12) + 2\log 2$, and ψ is increasing. With $y = h+\tau$ this shows that the difference we subtract is more than $2\log 2$. For $x>0$, [40, 5.9.15] writes $\log x - \frac{1}{2x} - \psi(x)$ as $2\int_0^\infty \frac{t\,dt}{(t^2+x^2)(e^{2\pi t}-1)}$, which lies in $(0, \frac{1}{12x^2})$ since $\int_0^\infty \frac{t\,dt}{e^{2\pi t}-1} = \frac{1}{24}$. So $\psi(\Sigma) < \log\Sigma - \frac{1}{2\Sigma}$, and $\psi(x) > \log x - \frac{1}{2x} - \frac{1}{12x^2}$ for $x = X$ and $x = Y$. Hence the derivative is less than $\log\frac{\Sigma^2}{4XY} + (\frac{1}{2X} + \frac{1}{2Y} - \frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2} + \frac{1}{Y^2})$. Now $X - Y = 2b - s$, so $4XY = \Sigma^2 - (X-Y)^2 = \Sigma^2(1-z)$ and $X^2+Y^2 = \Sigma^2(1+z)/2$. So the three terms are $-\log(1-z)$, $\frac{X^2+Y^2}{2XY\Sigma} = \frac{1+z}{(1-z)\Sigma}$ and $\frac{X^2+Y^2}{12X^2Y^2} = \frac{2(1+z)}{3\Sigma^2(1-z)^2}$. □

Proposition A.4. *Let $0<\varepsilon<\frac14$, $\kappa = \frac{2\varepsilon}{1-4\varepsilon}$, $\kappa_s = \kappa(s+10)$ and $\Sigma_s = s + 2\kappa_s$. For $1 \le s \le N-1$ and $b \in \mathcal{S}_s$ let $\Delta_s(b) = \mathcal{L}_s\Phi_{s+1}^{\kappa_{s+1}}(b)\big/\Phi_s^{\kappa_s}(b)$.*

(a) Let $c = \kappa_s$, $c' = \kappa_{s+1}$, $d = \frac{s}{2} - b$, $u = \frac{d}{s+2c'}$, $v = \frac{d}{N-s}$ and $\rho = \frac{s+2c'}{s(1+2\alpha)}$. Then $|u| < \frac12$ and

$$\Delta_s(b) = \mathcal{J}\,\frac{\Phi_s^{c'}(b)}{\Phi_s^{c}(b)}, \qquad \mathcal{J} = 1 - (\rho-1)\frac{u(u+v)}{\frac14 - u^2}.$$

(b) The factor satisfies $\log\Delta_s(b) \le \frac{\kappa}{\Sigma_s} + \frac{2\kappa}{3\Sigma_s^2}$.

(c) For $1 \le t \le h$, $V_{t+1}(1) \le \pi_t \exp\big(\kappa\big[H_{N-1} - H_t + \frac{2}{3t}\big] + \kappa_{t+1}\big[\log\frac{(t+1)^2}{4t} + \frac23\big]\big)$.

(d) The center satisfies $C_N \le U_N := \varepsilon c_1 e^{\kappa H_{N-1}} \sum_{t=1}^h w_t(1-\varepsilon)^{t-1} e^{\kappa\mathcal{G}(t)}$, where $\mathcal{G}(t) = (t+11)\big(\log\frac{(t+1)^2}{4t} + \frac23\big) - H_t + \frac{2}{3t}$.

The slope κ solves $2(\kappa-\alpha)/(1+2\alpha) = \kappa$, so that in (b) the gain in $\mathcal{J}$ cancels the main loss from raising the extra balls from c to c'.

Proof. (a) Write $r' = r_s^{c'}(b)$, $\bar r = \bar r_s(b)$ and $r = r_s(b)$. By Lemma A.1(b) with c', and the recursion in Lemma A.1(a), we have $r'\Phi_{s+1}^{c'}(b+1) = \bar r\,\Phi_s^{c'}(b)$ and $(1-r')\Phi_{s+1}^{c'}(b) = (1-\bar r)\Phi_s^{c'}(b)$. So the numerator of $\Delta_s(b)$ divided by $\Phi_s^{c'}(b)$ is $\frac{(1-r)(1-\bar r)}{1-r'} + \frac{r\bar r}{r'}$. Here $r' = \frac12 - u$, $\bar r = \frac12 + v$ and $r = \frac12 - \rho u$, since $\frac12 - r = (1-2\varepsilon)\frac{d}{s} = \rho u$. Over the denominator $r'(1-r') = \frac14 - u^2$ the numerator is $f(u,v) + f(-u,-v)$ with $f(u,v) = (\frac12+\rho u)(\frac12 - v)(\frac12 - u)$, and this equals $\frac14 - \rho u^2 - (\rho-1)uv$. Also $|d| \le \frac{s}{2} - 1$, so $|u| < \frac12$.

(b) *Step 1. The factor $\mathcal{J}$.* Write $c = \kappa_s$, $c' = c + \kappa$ and $\Sigma = \Sigma_s$. Since $c' - \alpha s = (\kappa-\alpha)s + 11\kappa$, the choice of κ gives $\rho - 1 = \kappa + \nu_s$ with $\nu_s = \frac{22\kappa}{s(1+2\alpha)}$. As u and v have the sign of d, $u(u+v) \ge u^2$. So $\mathcal{J} \le 1 - (\kappa+\nu_s)W'$ with $z' = 4u^2$ and $W' = \frac{z'}{1-z'}$. Also $\mathcal{J} > 0$ by (a), since $0 < r < 1$ and $\bar r$, $1-\bar r$ are not both 0. So $\log\mathcal{J} \le -(\kappa+\nu_s)W'$.

Step 2. The ratio $\Phi_s^{c'}/\Phi_s^c$. The logarithm $\log(\Phi_s^{c'}(b)/\Phi_s^c(b))$ is the integral over $\tau \in [c, c']$ of the derivative in Lemma A.3. Its bound increases in z and decreases in Σ, while z decreases and Σ increases in τ. So we may take the bound at $\tau = c$, where $X + Y = \Sigma$, and multiply it by the length κ of the interval. Put $W = \frac{z}{1-z}$. Then $\frac{1}{1-z} = 1 + W$, $\frac{1+z}{1-z} = 1 + 2W$ and $-\log(1-z) \le W$, so

$$\log\Delta_s(b) \le \kappa\Big[W + \frac{1+2W}{\Sigma} + \frac{2(1+2W)(1+W)}{3\Sigma^2}\Big] - (\kappa+\nu_s)W'. \tag{3}$$

Step 3. What is left to show. The terms of (3) without W and W' are $\frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2}$, which is the bound we want. Since $(1+2W)(1+W) = 1 + 3W + 2W^2$, the other terms are

$$\kappa(W - W') + \frac{2\kappa W}{\Sigma} + \frac{2\kappa W}{\Sigma^2} + \frac{4\kappa W^2}{3\Sigma^2} - \nu_s W',$$

and we must show that they add up to at most 0. If $W = 0$ then $d = 0$, so $W' = 0$ and there is nothing to do. So let $W > 0$. The first four terms are positive, and we show that each is a small multiple of $\frac{\kappa}{s}W$. The last term is negative (it comes from the shift 10 in κ_s), and it is a larger multiple of $\frac{\kappa}{s}W$. We need three facts, with $X = b + c$ and $Y = s - b + c$ as in Step 2.

(i) We have $1 + W \le \frac{\Sigma}{2(1+c)}$. Indeed $X, Y \ge 1 + c$, since $b \ge 1$ and $s - b \ge 1$, and the larger of X and Y is at least $\frac{\Sigma}{2}$. So $XY \ge (1 + c)\Sigma/2$, and $1 + W = \frac{1}{1-z} = \frac{\Sigma^2}{4XY}$.

(ii) We have $W - W' \le \frac{2\kappa}{1+c}W$, and so $W' \ge \frac{9}{11}W$. Indeed z and z' have the same numerator $(s-2b)^2$, and their denominators are Σ^2 and $(s+2c')^2 = (\Sigma+2\kappa)^2$. So $z' = \lambda z$ with $\lambda = (\frac{\Sigma}{\Sigma+2\kappa})^2 \ge 1 - \frac{4\kappa}{\Sigma}$. Then

$$W - W' = \frac{(1-\lambda)z}{(1-z)(1-\lambda z)} \le (1-\lambda)W(1+W) \le \frac{4\kappa}{\Sigma}\, W\, \frac{\Sigma}{2(1+c)} = \frac{2\kappa}{1+c}W,$$

where we used $1 - \lambda z \ge 1 - z$ and then (i). Since $c = \kappa(s + 10) \ge 11\kappa$, we have $\frac{2\kappa}{1+c} \le \frac{2}{11}$.

(iii) We have $\Sigma \ge s$, $1 + c > \kappa s$ and $\Sigma^2 \ge 3s$. The first two are clear. For the third, $W > 0$ means $s \ne 2b$, and with $b \ge 1$ and $s - b \ge 1$ this forces $s \ge 3$.

Step 4. The terms with W. We bound the four positive terms one by one.

$$\begin{aligned}
\kappa(W - W') &\le \frac{2\kappa^2}{1+c}W \le \frac{2\kappa}{s}W && \text{by (ii) and } 1 + c > \kappa s,\\
\frac{2\kappa W}{\Sigma} &\le \frac{2\kappa}{s}W && \text{by } \Sigma \ge s,\\
\frac{2\kappa W}{\Sigma^2} &\le \frac{2\kappa}{3s}W && \text{by } \Sigma^2 \ge 3s,\\
\frac{4\kappa W^2}{3\Sigma^2} &\le \frac{4\kappa W(1+W)}{3\Sigma^2} \le \frac{2\kappa W}{3\Sigma(1+c)} \le \frac{2\kappa}{3s}W && \text{by (i) and } \Sigma \ge s.
\end{aligned}$$

The four bounds add up to $\frac{16\kappa}{3s}W$. For the negative term, (ii) gives $-\nu_s W' \le -\frac{9}{11}\nu_s W = -\frac{18\kappa}{s(1+2\alpha)}W$. So

$$\log \Delta_s(b) \le \frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2} + W\Big[\frac{16\kappa}{3s} - \frac{18\kappa}{s(1+2\alpha)}\Big],$$

and the bracket is negative because $1 + 2\alpha < 2$ for $\varepsilon < \frac{1}{4}$, so $18/(1 + 2\alpha) > 9 > 16/3$.

(c) We first compare $V_{t+1}(1)$ with $\Phi_{t+1}^{\kappa_{t+1}}(1)$, and then this with $\pi_t = \Phi_{t+1}^{0}(1)$. For the first step apply Lemma A.2(a) with $s_0 = t + 1$, $F_s = \Phi_s^{\kappa_s}$ and $K_s = \exp(\frac{\kappa}{s} + \frac{2\kappa}{3s^2})$. This is allowed, since $F_N(h) = 1$, $F_s > 0$ on $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} = \Delta_s F_s \le K_s F_s$ on $\mathcal{S}_s$ by (b) and $\Sigma_s \ge s$. The product of the K_s over $t < s < N$ is the exponential of $\kappa(H_{N-1} - H_t) + \frac{2\kappa}{3}\sum_{s>t} s^{-2}$, and $\sum_{s>t} s^{-2} \le \frac{1}{t}$. Since $1 \in \mathcal{S}_{t+1}$, we get $V_{t+1}(1) \le e^{\kappa[H_{N-1} - H_t + 2/(3t)]}\, \Phi_{t+1}^{\kappa_{t+1}}(1)$.

For the second step, $\log(\Phi_{t+1}^{\kappa_{t+1}}(1)/\pi_t)$ is the integral over $\tau \in [0, \kappa_{t+1}]$ of the derivative in Lemma A.3 with $s = t+1$ and $b = 1$. As in Step 2 of (b), the bound of that lemma is largest at $\tau = 0$. There $X = 1$, $Y = t$ and $\frac{1}{1-z} = \frac{(t+1)^2}{4t}$, and the bound is $\log\frac{(t+1)^2}{4t} + \frac{t^2+1}{2t(t+1)} + \frac{t^2+1}{12t^2} \le \log\frac{(t+1)^2}{4t} + \frac{2}{3}$, because the two fractions are at most $\frac{1}{2}$ and $\frac{1}{6}$ for $t \ge 1$. Multiplying by the length κ_{t+1} of the interval gives (c).

(d) Insert (c) into Proposition 2.5(a), and use $\kappa_{t+1} = \kappa(t + 11)$ and $\pi_t = c_1 w_t$. □

Proof of Theorem 2.6. (a) By Proposition 2.5(a) and Lemma A.1(d), $C_N \ge \varepsilon c_1 \sum_t w_t(1 - \varepsilon)^{t-1}$. Now $(1-\varepsilon)^{t-1} \ge 1-(t-1)\varepsilon$ and $\sum_t w_t(t-1) = 2-\frac{6}{h+2}$ by Proposition 2.5(c). So $C_N \ge \varepsilon c_1(1-2\varepsilon)$, and $c_1 = \frac{4}{N+2}$.

(b) *Step 1. The constants.* Let $0 < \varepsilon(1+\log N) \le \frac{1}{100}$. Since $\log N \ge 0$, we have $\varepsilon \le \frac{1}{100}$. So $1-4\varepsilon \ge 0.96$ and $\kappa \le 2\varepsilon/0.96 \le 2.1\varepsilon$. Also $\kappa \le 2\varepsilon(1+8\varepsilon)$, because $(1+8\varepsilon)(1-4\varepsilon) = 1+4\varepsilon(1-8\varepsilon) \ge 1$. Since $\log \frac{(t+1)^2}{4t} \ge 0$ and $\frac{2}{3}(t+11) > 1+\log t \ge H_t$, we have $\mathcal{G}(t) > 0$. Since $\frac{(t+1)^2}{4t} \le \frac{t+1}{2}$, $\frac{2}{3} < \log 2$ and $H_t \ge \frac{2}{3t}$, also $\mathcal{G}(t) \le (t+11)\log(t+1) \le 12t\log(t+1)$. So $\kappa\mathcal{G}(t) \le 25.2\,\varepsilon t\log(t+1)$, and this is at most $\frac{t}{2}\log 2$ for $t \le h$, since then $25.2\,\varepsilon\log(t+1) \le 25.2\,\varepsilon\log N \le 0.252 < \frac{1}{2}\log 2$.

Step 2. The sum over t. Since $e^y - 1 \le ye^y$ for $y \ge 0$, Step 1 gives $e^{\kappa\mathcal{G}(t)} - 1 \le \kappa\mathcal{G}(t)2^{t/2}$. By Proposition A.4(d), $(1-\varepsilon)^{t-1} \le 1$, $\sum_t w_t = 1$ and $w_t \le 2t2^{-t}$ (Proposition 2.5(c)),

$$C_N \le \varepsilon c_1 e^{\kappa H_{N-1}}\Big(1 + 50.4\,\varepsilon\sum_{t\ge 1} t^2\log(t+1)2^{-t/2}\Big),$$

and we bound the series as follows. The ratio of its consecutive terms is $(1+\frac{1}{t})^2\frac{\log(t+2)}{\log(t+1)}2^{-1/2}$, which decreases in t and is below $\frac{3}{4}$ at $t = 60$. So the terms with $t > 60$ add up to less than four times the term at $t = 61$, and this is below 10^{-4}. The first 60 terms add up to 102.5864 to four decimals (enclosed in ball arithmetic by `certify_elephant.py`), so the series is below 102.59.

Step 3. The factor $e^{\kappa H_{N-1}}$. Finally $H_{N-1} \le 1+\log N$, so $\kappa H_{N-1} \le 2\varepsilon\log N + 2\varepsilon + 16\varepsilon^2(1+\log N) \le 2\varepsilon\log N + 3\varepsilon$. With $1+y \le e^y$ this proves (b), since $3 + 50.4\cdot 102.59 < 5200$. Any upper bound for the series below $(5200-3)/50.4$, which exceeds 103.1, would give the same constant. □

B The families and the mode

This appendix states the family estimates used for the mode bound in Corollary 3.11. Their proofs are in Appendix D, in the order stated here, and also support the tail theorem in the arXiv version of this paper [41]. We keep the conventions of Section 3. All constants in this appendix are explicit. Appendix D writes C for an absolute constant whose value may change from line to line. Since M is independent of X_1, $f_N(m) = \mathbb{P}(M_N = m)$. Every path of a chain $M^{[k_0]}$ is nondecreasing in time, and we use the paths of Y and $Z^{(j)}$ fixed after Lemma 3.3. The size $|\mathcal{T}_v|$ of the family has the law β_v of Lemma 2.4. Let $\bar\beta_v(s) = \mathbb{P}(|\mathcal{T}_v| \ge s)$, with $\bar\beta_v(s) = 1$ for $s \le 1$. By Pascal's rule $\binom{N-s}{v-1} - \binom{N-s-1}{v-1} = \binom{N-s-1}{v-2}$ the tail of β_v telescopes. So, for $1 \le s \le N-1$,

$$\bar\beta_v(s) = \binom{N-s}{v-1}\Big/\binom{N-1}{v-1} = \prod_{i=1}^{v-1}\Big(1 - \frac{s-1}{N-i}\Big) \le e^{-(v-1)(s-1)/(N-1)}, \qquad \beta_v(s) = \frac{v-1}{N-s}\,\bar\beta_v(s), \tag{4}$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{N-d}, \qquad \frac{\bar\beta_v(s-1)}{\bar\beta_v(s)} = 1 + \frac{v-1}{N-s-v+2}. \tag{5}$$

Lemma B.1 (monotone sandwich)**.** *Let $0 \le a \le b \le N$. If $m_h \le a$, then $(b-a+1)f_N(b) \le \mathbb{P}(a \le M_N \le b)$. If $b \le m_h$, then $(b-a+1)f_N(a) \le \mathbb{P}(a \le M_N \le b)$.*

Lemma B.2 (tail)**.** *Let $0 < \varepsilon < 1$, $k_0 \ge 1$ and $t \ge k_0$. For every integer $z \ge 1$,*

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \frac{\varepsilon t H_z}{z}, \qquad \mathbb{E}M_t^{[k_0]} \le \varepsilon t\log\frac{t}{k_0}.$$

In particular $\mathbb{P}(M_N \ge z) \le xH_z/z$, and $\mathbb{P}(Z_s^{(j)} \ge z) \le xH_z/z$ for $j-1 \le s \le N$.

Lemma B.3 (the first flipped family)**.** *For* $1 \le d \le N-1$,

$$\frac{x}{d(d+1)}\Big(1-\frac{2x(N-d-1)}{N(d+2)}\Big) \le \mathbb{P}\big(|\mathcal{T}_V| = d\big) \le \frac{x}{d(d+1)}, \qquad \mathbb{P}\big(|\mathcal{T}_V| \ge d\big) \le x\Big(\frac{1}{d}-\frac{1}{N}\Big).$$

Proposition B.4 (local lower bound)**.** *Let* $N \ge 64$, $0 < \varepsilon < 1$ *and* $2 \le m \le N/4$. *Then* $f_N(m) \ge \frac{x}{m(m+1)}(1-e_1'(m)) \ge \frac{x}{m(m+1)}(1-e_1(m))$, *where*

$$e_1'(m) = 2\varepsilon H_{m+1} + \frac{xH_m}{m} + \frac{2x}{m+2} + 8\varepsilon \log N + 20me^{-m/9}$$

and

$$e_1(m) = 20\Big(\frac{x\log(m+1)}{m} + \varepsilon \log N + me^{-m/9}\Big).$$

Idea of the proof. We use the formula of Lemma 3.3. Given the time j of the first flip, the count M_N is $d - Y_d + Z_{N-d}^{(j)}$, where d is the size of the first flipped family. As d grows by one this quantity moves by at most one, so it hits m at some d close to m unless Y or Z is large. (If no step of the family flips back and the rest of the tree has no flip, then $Y = Z = 0$ and the quantity is d itself, which hits m at $d = m$.) The law β_j of d is nearly flat near m, and this gives almost the full weight $\beta_j(m)$. Then we sum over j. The error e_1 collects the chance that Y or Z is large, the small slope of β_j, and the first flips that come too late to be treated this way.

Lemma B.5 (mode bound)**.** *Let* $N \ge 64$ *and* $x \ge 1$, *and put* $\bar{c} = \frac{1}{40}$ *and* $A_0 = 15400$. *If* $\varepsilon \log N \le \bar{c}$ *and* $804x\log(x+2) \le N$, *then* $m_h \le A_0 x \log^2(x+1)$. *In particular this holds if* $\varepsilon \log N \le \bar{c}$ *and* $A_0 x \log(x+2) \le N/4$.

C Proofs for the bulk

We prove Theorems 3.8 and 3.9, Lemma 3.10 and Corollary 3.11. We keep the notation of Section 3 and of Appendix B. So $\mathcal{T}$ is the tree of Lemma 3.1, $\mathcal{T}_v$ is the family of v, and μ_v is the mark of v. By Lemma 2.4 and the discussion after Lemma 3.1, $|\mathcal{T}_v|$ has the law β_v, with mean N/v, and $\sum_{v=2}^{N} \beta_v(d) = N/(d(d+1))$ for $1 \le d \le N-1$. Let

$$\Psi_N(\theta) = \sum_{d=1}^{N-1} \frac{e^{i\theta d}-1}{d(d+1)}, \qquad \Psi_\infty(\theta) = \sum_{d \ge 1} \frac{e^{i\theta d}-1}{d(d+1)}.$$

The law CP_N has the characteristic function $\widehat{\mathrm{CP}}_N(\theta) = e^{x\Psi_N(\theta)}$.

C.1 Proof of Theorem 3.8

We use $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_k |P(k)-Q(k)|$. The case $N = 1$ is immediate. Assume $N \ge 2$ and $0 < \varepsilon < 1$. Put $Z_+ = \sum_{v=2}^{N} \mu_v |\mathcal{T}_v|$. Unless two marked vertices are strictly nested, $M_N = Z_+$. The family means therefore give

$$\mathbb{P}(M_N \ne Z_+) \le \varepsilon^2 \sum_{v=2}^{N} (N/v - 1) = \varepsilon^2 (NH_N - 2N + 1).$$

Given the tree, replace the independent Bernoulli marks by independent Poisson variables of mean ε. The scalar total variation is $\varepsilon(1-e^{-\varepsilon}) \le \varepsilon^2$, since the only positive Bernoulli-minus-Poisson

mass is at 1. Product coupling and the weighted sum cost at most $(N-1)\varepsilon^2$. The resulting conditional law is $\mathrm{CP}(\Lambda)$, where $\Lambda = \varepsilon \sum_{v=2}^{N} \delta_{|\mathcal{T}_v|}$ has deterministic mass $\lambda = (N-1)\varepsilon$. By the family identity, $\mathbb{E}\Lambda\{d\} = x/[d(d+1)]$, so $\mathrm{CP}(\mathbb{E}\Lambda) = \mathrm{CP}_N$.

Use the L1 norm on signed measures and convolution $*$, for which $\|\sigma * \tau\|_1 \le \|\sigma\|_1 \|\tau\|_1$. On intensities of mass λ, the CP map and its derivatives in a zero-mass direction D are

$$F(\nu) = e^{-\lambda} \sum_{r \ge 0} \frac{\nu^{*r}}{r!}, \qquad F'(\nu)[D] = F(\nu) * D, \qquad F''(\nu)[D, D] = F(\nu) * D * D,$$

where $\nu^{*0} = \delta_0$. The series and derivatives converge in L1. Let $\mathcal{F}_j = \sigma(U_1, \ldots, U_j)$, $\Lambda_j = \mathbb{E}(\Lambda \mid \mathcal{F}_j)$ and $D_j = \Lambda_j - \Lambda_{j-1}$. Each Λ_j is nonnegative of mass λ. Along the segment between consecutive intensities F has norm 1, so Taylor's remainder has norm at most $\|D_j\|_1^2/2$. The linear term has conditional mean zero given $\mathcal{F}_{j-1}$. Telescoping expectations and dividing by two gives

$$\mathrm{TV}(\mathbb{E}F(\Lambda), F(\mathbb{E}\Lambda)) \le \frac{1}{4} \sum_{j=1}^{N-1} \mathbb{E}\|D_j\|_1^2.$$

If $\Lambda^{(j)}$ resamples only U_j, independence gives $D_j = \mathbb{E}(\Lambda - \Lambda^{(j)} \mid \mathcal{F}_j)$. Conditional Jensen thus bounds its squared norm in expectation by that of $\Lambda - \Lambda^{(j)}$. Moving the subtree of $j+1$ changes family sizes only along the old and new parent paths, excluding the root. Each altered atom contributes at most 2ε in L1. Writing dp for depth, and using $(a+b)^2 \le 2(a^2+b^2)$, gives

$$\mathbb{E}\|\Lambda - \Lambda^{(j)}\|_1^2 \le 16\varepsilon^2 \mathbb{E}\,\mathrm{dp}(U_j)^2 \le 16\varepsilon^2 (H_{j-1}^2 + H_{j-1}).$$

For the last inequality, the depth pgf satisfies $g_1(s) = 1$ and $j g_{j+1}(s) = s \sum_{v \le j} g_v(s)$. Subtraction gives $g_v(s) = \prod_{i=1}^{v-1} (i - 1 + s)/i$. Thus the depth has the law of a sum of independent Bernoulli$(1/i)$ variables, and its second moment is at most $H_{v-1}^2 + H_{v-1}$. Average over the independent uniform parent $U_j \in \{1, \ldots, j\}$.

Adding the three errors proves the stronger bound

$$\mathrm{TV}(\mathcal{L}(M_N), \mathrm{CP}_N) \le \varepsilon^2 \left\{ N H_N - N + 4 \sum_{j=0}^{N-2} (H_j^2 + H_j) \right\} \le \delta_\Sigma.$$

Finally $H_N \le 1 + \log N$ and $H_j \le H_N$ give $\delta_\Sigma \le (x^2/N)\{\log N + 2 + 4(H_N^2 + H_N)\} \le \delta_B$. Every singleton discrepancy is at most total variation. □

C.2 Proof of Theorem 3.9

We write $B(t) = -it \log|t| - \frac{\pi}{2}|t|$ for real $t \ne 0$, and $B(0) = 0$. For $t > 0$ this is the value of $u \log u$ at $u = -it$, with the principal logarithm, and $B(-t) = \overline{B(t)}$. We need three facts about Ψ_N and Ψ_∞.

Fact 1. For real θ, $|\Psi_N(\theta) - \Psi_\infty(\theta)| \le \sum_{d \ge N} \frac{2}{d(d+1)} = \frac{2}{N}$.

Fact 2. For $0 < |\theta| \le \pi$ we have $\Psi_\infty(\theta) = -i\theta \log(-i\theta) + R(\theta)$ with $|R(\theta)| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big)$. The reason is that Ψ_∞ has a closed form in which θ enters only through $1 - e^{i\theta}$, and $1 - e^{i\theta}$ is close to $-i\theta$ for small θ. The term R is what this replacement changes. We prove it in the next four paragraphs.

We start with the closed form. For $0 < |s| \le 1$, $s \ne 1$,

$$\sum_{d\ge1} \frac{s^d-1}{d(d+1)} = \chi(s) := \frac{(1-s)\log(1-s)}{s},$$

because $\frac{1}{d(d+1)} = \frac{1}{d} - \frac{1}{d+1}$, $\sum_d s^d/d = -\log(1-s)$, $\sum_d s^d/(d+1) = (-\log(1-s)-s)/s$ and $\sum_d \frac{1}{d(d+1)} = 1$. So $\Psi_\infty(\theta) = \chi(e^{i\theta})$.

Next we compare $1-e^{i\theta}$ with $-i\theta$. Write $1-e^{i\theta} = (-i\theta)\,\mathsf{u}$ with $\mathsf{u} = \operatorname{sinc}(\theta/2)\,e^{i\theta/2}$, where $\operatorname{sinc} y = \sin y/y$ is positive for $|y| \le \frac{\pi}{2}$. So u is close to 1 when θ is small. The argument of $-i\theta$ is $\mp\frac{\pi}{2}$ for $\pm\theta > 0$ and that of u is $\theta/2$. They add up to a number in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\log(1-e^{i\theta}) = \log(-i\theta) + \log \mathsf{u}$ for principal logarithms.

Now we get a formula for R. By the two steps above, $\chi(e^{i\theta}) = (-i\theta)\,\mathsf{u}e^{-i\theta}\big(\log(-i\theta) + \log \mathsf{u}\big)$. Subtracting $-i\theta\log(-i\theta)$ gives

$$R(\theta) = -i\theta\log(-i\theta)\big(\mathsf{u}e^{-i\theta}-1\big) - i\theta\,\mathsf{u}e^{-i\theta}\log \mathsf{u}.$$

It remains to bound the three quantities in this formula. We show that

$$|\mathsf{u}e^{-i\theta}-1| \le 0.631\,|\theta|, \qquad |\log \mathsf{u}| \le 0.53\,|\theta|, \qquad |\log(-i\theta)| \le |\log|\theta|| + \frac{\pi}{2}.$$

For the first, put $y = \theta/2$. Then $|\mathsf{u}e^{-i\theta}-1| = |\operatorname{sinc} y - e^{iy}| \le (1-\operatorname{sinc} y) + |1-e^{iy}| \le \frac{y^2}{6} + |y| \le (\frac{1}{2} + \frac{\pi}{24})|\theta|$, where the last step uses $|\theta| \le \pi$, and $\frac{1}{2} + \frac{\pi}{24} \le 0.631$. For the second, $|\log \mathsf{u}|^2 = \log^2 \operatorname{sinc}(\theta/2) + \theta^2/4$, and $-\log\operatorname{sinc} y = \sum_{k\ge1} \frac{\zeta(2k)}{k\pi^{2k}} y^{2k}$ has positive coefficients. So $|\log\operatorname{sinc}(\theta/2)|/|\theta|$ increases in $|\theta|$, and its value at $|\theta| = \pi$ gives $|\log \mathsf{u}| \le \big(\frac{1}{4} + \frac{\log^2(\pi/2)}{\pi^2}\big)^{1/2}|\theta| \le 0.53\,|\theta|$. The third holds because $-i\theta$ has modulus $|\theta|$ and argument $\pm\frac{\pi}{2}$. Since $|\mathsf{u}e^{-i\theta}| \le 1$, the formula for R now gives

$$|R(\theta)| \le |\theta|\Big(|\log|\theta|| + \frac{\pi}{2}\Big)\cdot 0.631\,|\theta| + |\theta|\cdot 0.53\,|\theta| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big),$$

as $0.631\cdot\frac{\pi}{2} + 0.53 \le 1.53$.

Fact 3. For $|\theta| \le \pi$, $-\operatorname{Re}\Psi_\infty(\theta) = \sum_d \frac{1-\cos\theta d}{d(d+1)} \ge \frac{|\theta|}{2\pi}$. We may take $\theta \ne 0$. Keep only the terms with $d \le D := \lfloor \pi/|\theta| \rfloor$ and use $1-\cos v \ge 2v^2/\pi^2$ for $|v| \le \pi$. This gives at least $\frac{2\theta^2}{\pi^2}\sum_{d\le D}\frac{d}{d+1} \ge \frac{\theta^2 D}{\pi^2} \ge \frac{|\theta|}{2\pi}$, since $D \ge \frac{\pi}{2|\theta|}$.

The function e^B is a characteristic function of class L. Let S_x have the law CP_∞ with parameter x, namely the law of $\sum_{d\ge1} d\,\mathcal{N}_d$. Its characteristic function is $e^{x\Psi_\infty(\theta)}$. By Fact 2, for $t \ne 0$ and $x \ge \max(1, |t|/\pi)$,

$$x\Psi_\infty(t/x) - it\log x = B(t) + xR(t/x), \qquad |xR(t/x)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big). \quad (6)$$

So the characteristic function of $(S_x - x\log x)/x$ tends to $e^{B(t)}$ as $x \to \infty$, for every real t. The limit is continuous at 0, so by Lévy's continuity theorem it is the characteristic function of a probability law. It is integrable, so this law has the bounded continuous density f_L of Theorem 3.9. For $0 < c < 1$ and real t we have $B(ct) = cB(t) - ict\log c$, and hence

$$e^{B(t)} = e^{B(ct)}\cdot e^{B((1-c)t)}\,e^{it[(1-c)\log(1-c) + c\log c]}.$$

The last two factors form the characteristic function of a scaled and shifted copy of the law. So the law is of class L (self-decomposable), and f_L is unimodal by [33].

Proof of (a). By Fourier inversion on $\mathbb{Z}$ and the substitution $\theta = t/x$,

$$x\,\mathrm{CP}_N(m) = \frac{1}{2\pi}\int_{-\pi x}^{\pi x} e^{-i\lambda_m t} e^{A(t)}\,dt, \qquad A(t) = x\Psi_N(t/x) - it\log x.$$

Let $|t| \le \pi x$. By (6) and Fact 1,

$$|A(t) - B(t)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big) + \frac{2x}{N},$$

and by Facts 1 and 3, $\max(\mathrm{Re}\,A(t), \mathrm{Re}\,B(t)) \le -\frac{|t|}{2\pi} + \frac{2x}{N}$. Now $|e^A - e^B| \le |A-B|\,e^{\max(\mathrm{Re}\,A, \mathrm{Re}\,B)}$, and $\frac{2x}{N} \le \frac{1}{2}$ because $N \ge 4x$. So for $|t| \le \pi x$

$$|e^{A(t)} - e^{B(t)}| \le e^{1/2} e^{-|t|/2\pi}\Big[\frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big) + \frac{2x}{N}\Big].$$

For $s \ge 1$ the tangent of the logarithm at 16 gives $\log s \le s/16 + \log 16 - 1$, and for $0 < s < 1$ we have $s|\log s| \le 1/e$. So $s^2|\log s| \le s^3/16 + (4\log 2 - 1)s^2 + s/e$ for every $s \ge 0$. We insert this with $s = |t|$ and integrate, using $\int_0^\infty t^k e^{-t/2\pi}\,dt = k!\,(2\pi)^{k+1}$. This gives

$$\frac{1}{2\pi}\int_{-\pi x}^{\pi x} |e^{A(t)} - e^{B(t)}|\,dt \le e^{1/2}\Big[\frac{K_0 + 16\pi^2 \cdot 0.631\log x}{x} + \frac{4x}{N}\Big],$$

where $K_0 = 6\pi^3 \cdot 0.631 + 16\pi^2\big(0.631(4\log 2 - 1) + 1.53\big) + 4\pi \cdot 0.631/e \le 538.6$. Beyond πx we use $|e^{B(t)}| = e^{-\pi|t|/2}$, which gives $\frac{1}{2\pi}\int_{|t|>\pi x} |e^{B(t)}|\,dt = \frac{2}{\pi^2} e^{-\pi^2 x/2} \le \frac{4}{\pi^4 x} \le \frac{0.042}{x}$, since $e^{-y} \le 1/y$ for $y > 0$. Since $f_L(\lambda_m) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-i\lambda_m t} e^{B(t)}\,dt$, the error is at most the sum of the two integrals. With $e^{1/2} \le 1.6488$ and $16\pi^2 \cdot 0.631 \le 99.66$ this sum is at most

$$\frac{1.6488 \cdot 538.6 + 0.042 + 1.6488 \cdot 99.66\log x}{x} + \frac{6.5952\,x}{N} \le \frac{889 + 165\log x}{x} + \frac{6.6\,x}{N}.$$

The second bound in (a) holds because $165 \le 889$. This proves (a). Fourier inversion on $\mathbb{Z}$ holds at every integer m, and $\mathrm{CP}_N(m) = 0$ for $m < 0$, so the proof covers every integer m.

Proof of (b). Since $\varepsilon \le \frac{1}{4}$, we have $N \ge 4x$. By Theorem 3.8, $|x f_N(m) - x\,\mathrm{CP}_N(m)| \le x\delta_\Sigma \le x\delta_B$. Add (a).

Proof of (c). We drop the index n. Let $x \to \infty$ and $x^3 \log^2 N/N \to 0$. Then $N \ge x \to \infty$, so $\log N \ge 1$ for large n, and $x/N \to 0$. Hence for large n we have $\varepsilon \le \frac{1}{4}$ and $x \ge 1$, and (b) applies. Its bound tends to 0, since $x\delta_B = O(x^3 \log^2 N/N)$. Only large n matter for convergence in law. Let U be uniform on $[0,1)$ and independent of M_N. Then $(M_N + U - x\log x)/x$ has the density q_N with $q_N(\lambda) = x f_N(m)$ for $\lambda_m \le \lambda < \lambda_m + \frac{1}{x}$. On a compact set of λ the relevant m are nonnegative once x is large, and $|\lambda - \lambda_m| < \frac{1}{x}$. As f_L is uniformly continuous, $q_N \to f_L$ uniformly on compact sets. By Scheffé's lemma $q_N \to f_L$ in L^1, and since $U/x \to 0$ this gives the convergence in law. At the crossing $x_N \sim 2\log N$ by Theorem 2.7, so both conditions hold. □

C.3 Proof of Lemma 3.10

Step 1. A real integral. We claim that for every real λ and $k \ge 0$,

$$f_L^{(k)}(\lambda) = \frac{1}{\pi}\int_0^\infty (-s)^k\, e^{-s\log s - \lambda s}\sin(\pi s)\,ds. \tag{7}$$

For $k=0$ this is [24, Eq. (4.3)], and we include a short proof. Since $B(-t)=\overline{B(t)}$, the definition of f_L gives $f_L(\lambda)=\frac{1}{\pi}\operatorname{Re}\int_0^\infty \Phi(-it)\,dt$ with $\Phi(u)=\exp(u\log u+\lambda u)$. The function Φ is holomorphic off the negative real axis. Put $u=\rho e^{i\varphi}$ with $-\pi<\varphi\le -\frac{\pi}{2}$. Then $\operatorname{Re}(u\log u)=\rho(\log\rho\cos\varphi-\varphi\sin\varphi)$, where $\cos\varphi\le 0$ and $\varphi\sin\varphi\ge 0$. For $\log\rho\ge|\lambda|+1$ this gives $\operatorname{Re}(u\log u+\lambda u)\le -\rho(|\cos\varphi|+\varphi\sin\varphi)\le -\rho$, because $|\cos\varphi|+\varphi\sin\varphi\ge 1$ on $[-\pi,-\frac{\pi}{2}]$ (for $\psi=\varphi+\pi\in[0,\frac{\pi}{2}]$ it equals $\cos\psi+(\pi-\psi)\sin\psi\ge\cos\psi+\sin\psi\ge 1$). Also Φ is bounded near 0. So by Cauchy's theorem we may turn the ray $u=-it$ to the lower side of the negative real axis, where $u=-s$ and $\log u=\log s-i\pi$. There $u\log u=-s\log s+i\pi s$, and $dt=i\,du=-i\,ds$. Hence

$$f_L(\lambda)=\frac{1}{\pi}\operatorname{Re}\Big[-i\int_0^\infty e^{-s\log s-\lambda s}\,e^{i\pi s}\,ds\Big]=\frac{1}{\pi}\int_0^\infty e^{-s\log s-\lambda s}\sin(\pi s)\,ds.$$

The integrand decays faster than every exponential, so we may differentiate under the integral sign. This proves (7). Also f_L'' is bounded on the real line, because $t^2e^{B(t)}$ is integrable and we may differentiate the Fourier formula for f_L twice.

Step 2. Certified enclosures. Let $-0.223\le\lambda\le 0$ and $k\in\{1,2\}$, and split the integral in (7) at $\delta=2^{-20}$ and at 30. On $[0,\delta]$ we have $-s\log s\le 1/e$, $-\lambda s\le 0.223\,s$ and $|\sin\pi s|\le\pi s$, so this part is at most $\pi e^{1/e+0.223\delta}\delta^{k+2}/(k+2)$ in absolute value. On $[30,\infty)$ we have $s\log s+\lambda s\ge cs$ with $c=\log 30-0.223>3$, so this part is at most $\int_{30}^\infty s^2e^{-cs}\,ds\le 30^2e^{-30c}$. Both are below 10^{-17}. On $[\delta,30]$ the integrand is holomorphic in a neighborhood of the path, and we enclose the integral with the rigorous integration algorithm of [34], in ball arithmetic with 128 bits. The parameter λ enters as a ball, so one run encloses the integral for every λ in an interval. The script `certify_elephant.py` does this and reports

$$f_L'(-0.2229)\in[9.232\cdot10^{-6},9.234\cdot10^{-6}],\qquad f_L'(-0.2227)\in[-6.548\cdot10^{-6},-6.546\cdot10^{-6}],$$

and $f_L''(\lambda)\in[-0.0792,-0.0786]$ for every $\lambda\in[-0.2229,-0.2227]$. In the same way it finds that f_L' is positive at -0.22278299 and negative at -0.22278297. The two enclosures are $6.8984026\cdot10^{-10}$ and $-8.8809315\cdot10^{-10}$ with radii below 10^{-17}, and these are the smallest margins in the proof.

Step 3. The mode. Write $[a,b]=[-0.2229,-0.2227]$. By Step 2, f_L' is strictly decreasing on $[a,b]$ with $f_L'(a)>0>f_L'(b)$. So f_L' has exactly one zero λ_{L} in $[a,b]$, and f_L is strictly increasing on $[a,\lambda_{\mathrm{L}}]$ and strictly decreasing on $[\lambda_{\mathrm{L}},b]$. Since f_L is unimodal, there is a point λ_* such that f_L is nondecreasing on $(-\infty,\lambda_*]$ and nonincreasing on $[\lambda_*,\infty)$. If $\lambda_*>\lambda_{\mathrm{L}}$, then f_L would be nondecreasing just to the right of λ_{L}, which is false. In the same way $\lambda_*<\lambda_{\mathrm{L}}$ is impossible. So $\lambda_*=\lambda_{\mathrm{L}}$, and f_L is nondecreasing on $(-\infty,\lambda_{\mathrm{L}}]$ and nonincreasing on $[\lambda_{\mathrm{L}},\infty)$. By Step 2 again, $\lambda_{\mathrm{L}}\in(-0.22278299,-0.22278297)$. This proves (a) and (b).

Step 4. The constant κ_L. Let $\delta_0=\min(\lambda_{\mathrm{L}}-a,b-\lambda_{\mathrm{L}})>0$. For $|\lambda-\lambda_{\mathrm{L}}|\le\delta_0$, Taylor's formula with $f_L'(\lambda_{\mathrm{L}})=0$ and $f_L''\le -0.078$ on $[a,b]$ gives $f_L(\lambda_{\mathrm{L}})-f_L(\lambda)\ge 0.039(\lambda-\lambda_{\mathrm{L}})^2$. For $|\lambda-\lambda_{\mathrm{L}}|>\delta_0$ the monotonicity of Step 3 gives $f_L(\lambda)\le f_L(\lambda_{\mathrm{L}}\pm\delta_0)\le f_L(\lambda_{\mathrm{L}})-0.039\,\delta_0^2$. Since $\delta_0<1$, the bound in (c) holds with $0.039\,\delta_0^2$ in place of κ_L. By (a), $\lambda_{\mathrm{L}}-a>0.2229-0.22278299>1.17\cdot10^{-4}$ and $b-\lambda_{\mathrm{L}}>0.22278297-0.2227=8.297\cdot10^{-5}$, so $0.039\,\delta_0^2>0.039\cdot(8.297\cdot10^{-5})^2>2.68\cdot10^{-10}$. So (c) holds with $\kappa_L=2.6\cdot10^{-10}$. □

C.4 Proof of Corollary 3.11

The method is to compare f_N at a mode with f_N at the integer m' nearest to the predicted place $x\log x+\lambda_{\mathrm{L}}x$. By Theorem 3.9(b) both values are close to values of f_L, and f_L falls off quadratically away from λ_{L}. So a mode cannot be far from m'.

We first prove (a). So we assume (H), $E' < \kappa_L/2$ and $m_h \le N/4$, with E_1 and E' as in the statement.

Let m' be the integer nearest to $x\log x + \lambda_{\mathrm{L}} x$. This number increases in $x \ge 1$ and is at least $\lambda_{\mathrm{L}} > -\frac{1}{2}$, so $m' \ge 0$. Also $|\lambda_{m'} - \lambda_{\mathrm{L}}| \le \frac{1}{2x}$, and since $f_L'(\lambda_{\mathrm{L}}) = 0$, $f_L(\lambda_{m'}) \ge f_L(\lambda_{\mathrm{L}}) - \|f_L''\|_\infty/(8x^2)$. Let $\tilde m$ be a mode of f_N or $\tilde m = m^*$. We claim that $f_N(\tilde m) \ge f_N(m') - 16xH_N/N^2$. For a mode of f_N this is clear. For m^* we have $f_N(m^*) + f_N(N-m^*) = 2P_N(N-m^*) \ge 2P_N(N-m') \ge f_N(m')$. Here $N - m^* \ge \lceil N/2\rceil \ge \lceil N/4\rceil \ge m_h$. So f_N is nonincreasing on $\{\lceil N/4\rceil, \dots, N\}$ by Lemma 3.4. There are at least $N/4$ integers from $\lceil N/4\rceil$ to $\lceil N/2\rceil$. So Lemma B.1 (with $a = \lceil N/4\rceil \ge m_h$ and $b = \lceil N/2\rceil$) gives the second inequality below, and Lemma B.2 gives the third,

$$f_N(N-m^*) \le f_N(\lceil N/2\rceil) \le \frac{\mathbb{P}(M_N \ge \lceil N/4\rceil)}{N/4} \le \frac{16xH_N}{N^2}.$$

Now Theorem 3.9(b), used at $\tilde m$ and at m', gives

$$f_L(\lambda_{\tilde m}) \ge x f_N(\tilde m) - E_1 \ge x f_N(m') - \frac{16x^2H_N}{N^2} - E_1 \ge f_L(\lambda_{m'}) - 2E_1 - \frac{16x^2H_N}{N^2} \ge f_L(\lambda_{\mathrm{L}}) - 2E'.$$

Since $2E' < \kappa_L$, Lemma 3.10(c) gives $(\lambda_{\tilde m} - \lambda_{\mathrm{L}})^2 \le 2E'/\kappa_L$, which is the claim.

Now we prove (b). Let N be large and $\log N \le x \le 2\log N$. Then $x\delta_B$, x/N, $16x^2H_N/N^2$ and $\|f_L''\|_\infty/(8x^2)$ are all at most $\log x/x$, because each is $O(\log^5 N/N)$ or $O(1/\log^2 N)$. So $E' \le C_1 \log x/x$ with an absolute constant C_1. As $\log x/x \le \log\log N/\log N$, for large N we have $E' < \kappa_L/2$. Also (H) holds, $\varepsilon \log N \le 2\log^2 N/N \le \frac{1}{40}$ and $804x\log(x+2) \le N$, so Lemma B.5 gives $m_h \le A_0 x \log^2(x+1) \le N/4$. Hence by (a) $|m^* - x\log x - \lambda_{\mathrm{L}} x| \le x\sqrt{2E'/\kappa_L} \le K\sqrt{x\log x}$ with $K = \sqrt{2C_1/\kappa_L}$, and the same holds for every mode of f_N. At the crossing $x_N = 2\log N - \log\log N + O(1)$ for even N, by Theorem 2.7(b). So $\log N \le x_N \le 2\log N$ for large even N. □

D Proofs for the family estimates

These proofs establish the estimates stated in Appendix B. We retain its notation, including the pathwise construction of the chains.

Proof of Lemma B.1. By Lemma 3.4, $f_N(\ell) \ge f_N(b)$ for $a \le \ell \le b$ in the first case, and $f_N(\ell) \ge f_N(a)$ in the second. Sum over ℓ. □

Proof of Lemma B.2. We use the tree of Lemma 3.1 on $[t]$, with marks only on $v > k_0$. Every odd vertex lies in the family of a marked vertex, so $M_t^{[k_0]} \le \sum_{v \text{ marked}} |\mathcal{T}_v|$. We call a family large if its size is at least z, and small otherwise. If $M_t^{[k_0]} \ge z$, then some marked family is large, or the small marked families have total size at least z. We use the union bound for the first event and Markov's inequality for the second. Marks and tree are independent, so

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \ge z) + \frac{\varepsilon}{z}\sum_v \mathbb{E}\big[|\mathcal{T}_v|; |\mathcal{T}_v| < z\big].$$

By the first sum in Lemma 2.4, with t in place of N, on average $\frac{t}{d(d+1)}$ vertices have a family of size d. So the first sum over v is $\sum_{d\ge z} \frac{t}{d(d+1)} \le \frac{t}{z}$, and the second is $\sum_{d=1}^{z-1} \frac{t}{d+1} = t(H_z - 1)$. Together they give $\frac{\varepsilon t H_z}{z}$. For the mean, only the vertices $v > k_0$ can be marked and $\mathbb{E}|\mathcal{T}_v| = t/v$, so $\mathbb{E}M_t^{[k_0]} \le \varepsilon \sum_{v=k_0+1}^{t} \frac{t}{v} \le \varepsilon t \log \frac{t}{k_0}$. □

Proof of Lemma B.3. By Lemma 3.3, $\mathbb{P}(|\mathcal{T}_V| = d) = \sum_j \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and $1-(j-2)\varepsilon \le (1-\varepsilon)^{j-2} \le 1$. Apply the two sums of Lemma 2.4, and sum the upper bound over $d' \ge d$. □

Proof of Proposition B.4. We bound the inner sum of Lemma 3.3 from below for each early first flip, namely for $2 \le j \le J := \lfloor N/8 \rfloor$. The terms with $j > J$ are nonnegative and we drop them.

Step 1. One value of j. Fix $2 \le j \le J$ and put $\Gamma_d = d - Y_d + Z^{(j)}_{N-d}$ for $1 \le d \le N-j+1$. The inner sum in Lemma 3.3 is $Q_j(m) = \mathbb{E}S_j$ with $S_j = \sum_d \beta_j(d)\mathbf{1}\{\Gamma_d = m\}$. Let $\mathcal{E}$ be the event $\{Z^{(j)}_{N-1} \le m-1,\ Y_{2m} \le m\}$. We claim that

$$S_j \ge \beta_j(m)\Big(\mathbf{1}_{\mathcal{E}} - \frac{2(j-2)}{N}\,Y_{2m}\Big).$$

Off $\mathcal{E}$ the right side is at most 0 and $S_j \ge 0$. So let $\mathcal{E}$ hold. As d grows by one, Y_d grows by 0 or 1 and $Z^{(j)}_{N-d}$ falls by 0 or 1. So every step of Γ lies in $\{-1,0,1\}$, and Γ cannot jump over m. It starts at $\Gamma_1 = 1 + Z^{(j)}_{N-1} \le m$ (note that $Y_1 = 0$). At $d^\circ = m + Y_{2m} \le 2m$ we have $\Gamma_{d^\circ} \ge d^\circ - Y_{d^\circ} \ge d^\circ - Y_{2m} = m$, since $Z^{(j)} \ge 0$ and Y is nondecreasing. (The index d° is allowed, because $2m + j \le N$.) So $\Gamma_d = m$ for some $d \le d^\circ$, and as β_j is nonincreasing, $S_j \ge \beta_j(d^\circ) = \beta_j(m + Y_{2m})$. It remains to compare $\beta_j(m+y)$ with $\beta_j(m)$ for $0 \le y \le m$. By (5),

$$\frac{\beta_j(m+y)}{\beta_j(m)} = \prod_{i=1}^{y}\Big(1 - \frac{j-2}{N-m-i}\Big) \ge 1 - \frac{2y(j-2)}{N}.$$

Here $N - m - i \ge N/2$, so each fraction is at most $\frac{2(j-2)}{N} \le \frac{1}{4}$, and a product of factors $1-q_i$ with $0 \le q_i \le 1$ is at least $1 - \sum_i q_i$. This proves the claim. Now $\mathbf{1}_{\mathcal{E}} \ge 1 - \mathbf{1}\{Y_{2m} > m\} - \mathbf{1}\{Z^{(j)}_{N-1} \ge m\}$. Taking means in the claim gives

$$Q_j(m) \ge \beta_j(m)\big[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z^{(j)}_{N-1} \ge m)\big] - \frac{2(j-2)}{N}\,\mathbb{E}Y_{2m}\,\beta_j(m).$$

Step 2. The chance that Y or Z is large. Lemma B.2 gives

$$\mathbb{P}(Y_{2m} > m) \le 2\varepsilon H_{m+1}, \qquad \mathbb{P}(Z^{(j)}_{N-1} \ge m) \le \frac{xH_m}{m}, \qquad \mathbb{E}Y_{2m} \le 2\varepsilon m \log N.$$

None of these bounds depends on j. Put $p = 2\varepsilon H_{m+1} + xH_m/m$. If $p > 1$, then $e_1'(m) > 1$ and there is nothing to prove. So let $p \le 1$.

Step 3. Sum over j. By Lemma 3.3, $f_N(m) \ge \sum_{j=2}^{J} \mathbb{P}(V=j)\,Q_j(m)$. With Steps 1 and 2 this gives

$$f_N(m) \ge (1-p)\sum_{j=2}^{J}\mathbb{P}(V=j)\,\beta_j(m) - \frac{4\varepsilon m \log N}{N}\sum_{j=2}^{J}\mathbb{P}(V=j)\,(j-2)\,\beta_j(m).$$

In the second sum we use $\mathbb{P}(V=j) \le \varepsilon$ and the second sum in Lemma 2.4. So the second term is at most

$$\frac{4\varepsilon m \log N}{N} \cdot \frac{2x(N-m-1)}{m(m+1)(m+2)} \le \frac{x}{m(m+1)}\,8\varepsilon \log N.$$

The first sum is $\mathbb{P}(|\mathcal{T}_V| = m)$ without the terms $j > J$. So by Lemma B.3 it is at least $\frac{x}{m(m+1)}(1 - \frac{2x}{m+2}) - \varepsilon \sum_{j>J}\beta_j(m)$.

Step 4. The late first flips $j > J$. A family that starts after time $N/8$ rarely reaches size m. By (4), $\beta_j(m) \le \frac{j-1}{N-m} r^{j-1}$ with $r = e^{-a}$ and $a = \frac{m-1}{N-1} \le \frac{1}{4}$. We sum this over

$j-1 \ge J$. With $\sum_{i\ge J} i r^i \le r^J\big(\frac{J}{1-r}+\frac{1}{(1-r)^2}\big)$, $1-e^{-a}\ge a/2$ and $N-m\ge \frac34 N$ we get $\varepsilon\sum_{j>J}\beta_j(m)\le \frac{4\varepsilon}{3N}\big(\frac{2J}{a}+\frac{4}{a^2}\big)e^{-aJ}$. Since $\lfloor N/8\rfloor \ge \frac{N-1}{9}$ for $N\ge 64$, we have $aJ\ge \frac{m-1}{9}$. Also $J\le N/8$ and $a\ge \frac{m-1}{N}$. So

$$\varepsilon\sum_{j>J}\beta_j(m)\le x\Big(\frac{1}{3(m-1)}+\frac{16}{3(m-1)^2}\Big)e^{-(m-1)/9}\le \frac{x}{m(m+1)}\,20me^{-m/9}.$$

The last step uses $m+1\le 3(m-1)$ and $m+1\le 3(m-1)^2$, which hold for $m\ge 2$, and $e^{1/9}\le\frac98$, since $17\cdot\frac98\le 20$.

Step 5. Collect. For $0\le p\le 1$ and $q\ge 0$ we have $(1-p)(1-q)\ge 1-p-q$. So Steps 3 and 4 give

$$f_N(m)\ge \frac{x}{m(m+1)}\Big(1-p-\frac{2x}{m+2}-20me^{-m/9}-8\varepsilon\log N\Big),$$

which is the bound with $e_1'(m)$. For $e_1'\le e_1$ we use $\log(m+1)\ge\log 3>1$ and $\log N\ge\log 64>4$. So $H_m\le 1+\log m\le 2\log(m+1)$, $H_{m+1}\le 1+\log N\le\frac54\log N$ and $\frac{2x}{m+2}\le\frac{2x\log(m+1)}{m}$. Hence $e_1'(m)\le\frac{4x\log(m+1)}{m}+\frac{21}{2}\varepsilon\log N+20me^{-m/9}\le e_1(m)$. □

Proof of Lemma B.5. We pick a point ℓ of order $x\log x$ at which $f_N(\ell)$ is at least half of $\frac{x}{\ell(\ell+1)}$. Up to its mode f_N is nondecreasing. So if the mode were far to the right of ℓ, the values from ℓ to the mode would carry more mass than the tail $\mathbb{P}(M_N\ge\ell)$ has.

Put $L_2=\log(x+2)\ge\log 3>1$ and $\ell=\lceil 200xL_2\rceil$, so that $200\le\ell\le 201xL_2$ and $4\ell\le 804xL_2\le N$. Since $\log N\ge 4$, we have $\varepsilon\le\frac{1}{160}$. We bound the five terms of $e_1'(\ell)$ in Proposition B.4. As $(1+\log s)/s$ decreases for $s\ge 1$, and $\log 200\le 6$, $\log x\le L_2$ and $\log L_2\le L_2$, we get $\frac{xH_\ell}{\ell}\le\frac{x(1+\log(200xL_2))}{200xL_2}\le\frac{7+2L_2}{200L_2}\le\frac{9}{200}$. Next $\frac{2x}{\ell+2}\le\frac{1}{100}$, $2\varepsilon H_{\ell+1}\le 2\varepsilon(1+\log N)\le\frac{1}{80}+\frac{1}{20}$ and $8\varepsilon\log N\le\frac15$. Finally $se^{-s/9}$ decreases for $s\ge 9$, so $20\ell e^{-\ell/9}\le 4000e^{-200/9}<10^{-5}$. The five terms add up to less than $\frac12$, so $f_N(\ell)\ge\frac{x}{2\ell(\ell+1)}$. If $m_h\ge\ell$, Lemmas B.1 (with $a=\ell$, $b=m_h$) and B.2 give $(m_h-\ell+1)f_N(\ell)\le\mathbb{P}(M_N\ge\ell)\le xH_\ell/\ell$. So $m_h\le(\ell+1)(1+2H_\ell)$ in every case. Here $\ell+1\le 202xL_2$, because $xL_2\ge 1$. Also $H_\ell\le 1+\log(201xL_2)\le 7+2L_2\le 9L_2$, so $1+2H_\ell\le 19L_2$. Hence $m_h\le 3838\,xL_2^2$. Since $(x+1)^2\ge x+2$, we have $L_2\le 2\log(x+1)$, and so $m_h\le 15352\,x\log^2(x+1)\le A_0x\log^2(x+1)$. The last claim holds because $804x\log(x+2)\le 4A_0x\log(x+2)$. □

E Proofs for the second difference

We keep the notation of Section 3. This section proves the results of Section 3.5 on the signs of the second difference.

E.1 Proof of Lemma 3.14

We insert the numbers η_i into ζ one at a time. If $\eta_i=0$, the sign pattern does not change. If $\eta_i>0$, then $\zeta_i>0$ or $\zeta_{i+1}>0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The case $\eta_i<0$ is the same. This proves the first claim. Deleting entries from a list replaces its sign pattern by a subsequence, and η is obtained from w by deleting the entries ζ_i. This proves the second claim. □

E.2 Proof of Lemma 3.15

Put $\xi_t = P_t(1)/P_t(0)$. By (1), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1-\varepsilon-\frac{c}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\xi_{t+1} - \xi_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{c\xi_t}{t}\Big), \qquad \xi_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\xi_t < \varepsilon t/c$ for $t \geq 2$. For $t = 2$ this holds because $c < 1-\varepsilon$. If it holds for t, then $0 < \xi_{t+1} - \xi_t \leq \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{c}$, and it holds for $t+1$. So ξ_t is strictly increasing for $t \geq 2$, and $\xi_t < 1$ whenever $t \leq c/\varepsilon$.

Now (2) with $a = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \geq 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{c}{t}\Big)(\xi_t - 2)\, P_t(0).$$

Here $1 - r_t(2) \geq 1 - \varepsilon - c = \varepsilon > 0$. If $t < c/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > c/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because ξ_t increases. So $\mathrm{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. Let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \geq \frac{1}{3}$, then $c/\varepsilon \leq 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-,+)$. This proves the first claim.

For the second claim let $0 \leq \varepsilon \leq 1$. The proofs of (1) and (2) use only that step $t+1$ is $+1$ with probability $r_t(a)$ given $A_t = a$, so both hold for this ε. We compute $P_3(0) = \frac{1}{2}(1-\varepsilon)^2$ and $P_3(1) = P_3(2) = \frac{1}{2} - P_3(0)$, so $s_3 = P_3(0) - P_3(1) = (1-\varepsilon)^2 - \frac{1}{2}$. Suppose that $s_3 = 0$. Then $\varepsilon = 1 - 2^{-1/2}$ and $P_3(0) = P_3(1) = \frac{1}{4}$. By (2) with $t = 3$ and $a = 1$, $s_4 = r_3(-1)D_3(0) = (\varepsilon - \frac{c}{3})(P_3(1) - 2P_3(0)) = -\frac{5\varepsilon - 1}{12}$. This is not 0, because $\varepsilon \neq \frac{1}{5}$. (With P_t defined by (1), the same computation works for every real ε, because the other root $1 + 2^{-1/2}$ of s_3 is not $\frac{1}{5}$ either. We need only $0 < \varepsilon < \frac{1}{2}$.) □

E.3 Proof of Theorem 3.16

For $3 \leq t \leq N$ let $\mathcal{Z}_t$ be the list

$$\mathcal{Z}_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N - t - 1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list $\mathcal{Z}_3$ is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and $\mathcal{Z}_N$ is the list in the statement.

We check that $\mathcal{Z}_{t+1}$ is obtained from $\mathcal{Z}_t$ as in Lemma 3.14, for $3 \leq t \leq N-1$. The first $N - t$ entries of $\mathcal{Z}_{t+1}$ are $s_N, \dots, s_{t+1}$, and they are copies of the first $N - t$ entries of $\mathcal{Z}_t$. So $A_i = 1$ and $B_i = 0$ there. The next $t - 2$ entries are $D_{t+1}(b)$ for $2 \leq b \leq t-1$. By (2),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big)D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of $\mathcal{Z}_t$, at the right places. The coefficients satisfy $r_t(b-2) \geq \varepsilon$ and $1 - r_t(b+1) \geq 1 - \varepsilon - c = \varepsilon$, since $2 \leq b \leq t-1$. The last $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_{t+1}, \ldots, s_N$, and each is a copy of the entry of $\mathcal{Z}_t$ one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 3.14, $\operatorname{sp}(\mathcal{Z}_N) \preceq \operatorname{sp}(\mathcal{Z}_{N-1}) \preceq \cdots \preceq \operatorname{sp}(\mathcal{Z}_3)$. By Lemma 3.15 the sign pattern p of $(s_3, \ldots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+,-)$, $(-,+)$ and $(+,-,+)$. The sign pattern of $\mathcal{Z}_3$ is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-,+,-)$, $(+,-,+)$ and $(+,-,+,-,+)$, and each is a subsequence of Π_5.

For f_N we argue in the same way with $D_t^f(m) = f_t(m+1) - 2f_t(m) + f_t(m-1)$ and the lists

$$\mathcal{Z}_t^f = \big(s_N^f, \ldots, s_{t+1}^f,\ s_t^f, D_t^f(2), \ldots, D_t^f(t-1),\ d_t, d_{t+1}, \ldots, d_{N-1}\big), \qquad 2 \leq t \leq N,$$

where $s_t^f = D_t^f(1)$ and $d_u = (1 - r_u(u+1))D_u^f(u) = (\varepsilon - \frac{c}{u})f_u(u-1)$. The entry $D_{t+1}^f(t)$ of $\mathcal{Z}_{t+1}^f$ equals $r_t(t-2)D_t^f(t-1) + d_t$ by (2), and the other entries are as before. The proof of Lemma 3.15 applies to s_t^f without change, with $\xi_t = f_t(1)/f_t(0)$, $\xi_2 = \frac{\varepsilon}{1-\varepsilon}$ and $s_2^f = 1 - 3\varepsilon$. The numbers d_u are negative for $u < c/\varepsilon$ and positive for $u > c/\varepsilon$. So $\operatorname{sp}(\mathcal{Z}_2^f)$ is a subsequence of $(+,-,+)$ followed by $(-,+)$, and $\mathcal{Z}_N^f = (D_N^f(1), \ldots, D_N^f(N-1))$. As before $\operatorname{sp}(\mathcal{Z}_N^f) \preceq \operatorname{sp}(\mathcal{Z}_2^f)$, and a subsequence of $(+,-,+)$ followed by $(-,+)$ is a subsequence of Π_5 once equal neighbors are merged. This argument needs only $N \geq 2$, and for $N = 2$ the lists $\mathcal{Z}_2^f$ and $\mathcal{Z}_N^f$ are the same. □

E.4 Proof of Theorem 3.17

By symmetry $D_N(N - a_1) > 0 > D_N(N - a_0)$. Suppose that $D_N(a) \leq 0$ for some a with $a_1 < a < N - a_1$.

The case $D_N(a) < 0$. Then the entries of $\mathcal{Z}_N$ at $a_0 < a_1 < a < N - a_1 < N - a_0$ have the signs $-,+,-,+,-$. So $(-,+,-,+,-) \preceq \operatorname{sp}(\mathcal{Z}_N) \preceq \Pi_5$ by Theorem 3.16. This is impossible, because a subsequence of Π_5 with five terms is Π_5 itself.

The case $D_N(a) = 0$. We use the lists $\mathcal{Z}_t$ of the proof of Theorem 3.16 and number the entries of $\mathcal{Z}_t$ from 1 to $2N - t - 1$. So $D_t(b)$ stands at place $N - t + b$ for $1 \leq b \leq t-1$. For $3 \leq t \leq N-1$ let W_t be the list w of Lemma 3.14 for $\zeta = \mathcal{Z}_t$ and $\eta = \mathcal{Z}_{t+1}$. Then $\operatorname{sp}(W_t) = \operatorname{sp}(\mathcal{Z}_t) \preceq \Pi_5$. The entry of $\mathcal{Z}_{t+1}$ at place k is a combination of the entries of $\mathcal{Z}_t$ at the places k and $k+1$, with nonnegative coefficients. We call these two entries its parents. For $2 \leq b \leq t-1$ the parents of $D_{t+1}(b)$ are $D_t(b-1)$ and $D_t(b)$, and both coefficients are positive.

For $4 \leq t \leq N$ let B_t be the set of integers b with $\max(2, a - N + t) \leq b \leq \min(a, t-2)$. It is not empty, because $2 \leq a \leq N-2$. We prove by downward induction on t that there are places $p_1 < p_2 < p_4 < p_5$ in $\mathcal{Z}_t$ such that the entries there have the signs $-,+,+,-$, that $D_t(b) = 0$ for every $b \in B_t$, and that all the places $N - t + b$ with $b \in B_t$ lie strictly between p_2 and p_4. For $t = N$ we have $B_N = \{a\}$, and we take the places a_0, a_1, $N - a_1$ and $N - a_0$.

Let the claim hold for $t+1$, where $4 \leq t \leq N-1$, with the places $p_1' < p_2' < p_4' < p_5'$. Let $b \in B_{t+1}$. Then $D_{t+1}(b) = 0$ is a combination of its two parents with positive coefficients. So the parents both vanish, or they have strictly opposite signs. In the second case one parent is negative. In W_t it stands next to $D_{t+1}(b)$, hence between the positive entries of $\mathcal{Z}_{t+1}$ at the places p_2' and p_4', and these stand between the negative entries at p_1' and p_5'. So $(-,+,-,+,-) \preceq \operatorname{sp}(W_t) \preceq \Pi_5$, which is impossible. Hence $D_t(b-1) = D_t(b) = 0$ for every $b \in B_{t+1}$. Every element of B_t is of the form b or $b-1$ with $b \in B_{t+1}$, so D_t vanishes on B_t.

The entry of $\mathcal{Z}_{t+1}$ at p_1' is negative, so one of its parents is negative. Let $p_1 \in \{p_1', p_1' + 1\}$ be its place in $\mathcal{Z}_t$, and choose p_2, p_4 and p_5 in the same way. Then $p_1 \leq p_1' + 1 \leq p_2' \leq p_2$, and

$p_1 \neq p_2$ because the signs differ. In the same way $p_4 < p_5$. Let q^- and q^+ be the smallest and the largest place $N - t - 1 + b$ with $b \in B_{t+1}$. The places $N - t + b$ with $b \in B_t$ lie between q^- and $q^+ + 1$. Since $p_2 \leq p_2' + 1 \leq q^-$ and $p_4 \geq p_4' \geq q^+ + 1$, and the entries at p_2 and p_4 are not zero, these places lie strictly between p_2 and p_4. This proves the claim for t.

The induction step with $t = 4$ shows that both parents of $D_5(b)$ vanish for every $b \in B_5 \subseteq \{2, 3\}$. The parents of $D_5(2)$ are s_4 and $D_4(2)$, and those of $D_5(3)$ are $D_4(2)$ and s_4. So $s_4 = D_4(2) = 0$. By (2) with $t = 3$ and $a = 2$, and since $D_3(1) = D_3(2) = s_3$, we have $D_4(2) = r_3(0)s_3 + \big(1 - r_3(3)\big)s_3 = 2\varepsilon s_3$. So $s_3 = 0$ as well, which contradicts Lemma 3.15. □

F Sign patterns of functions

We prove the lemmas used for Theorem 4.2, with the notation of Sections 3.5 and 4.2. Deleting entries from a list replaces its pattern by a subsequence. An alternating list p satisfies $p \preceq q$ for every alternating list q with the same first sign and at least as many entries.

Lemma F.1 (Descartes' rule for exponential sums)**.** *Let $\lambda_1 < \cdots < \lambda_n$ and $b_1, \ldots, b_n$ be real, with $b = (b_1, \ldots, b_n) \neq 0$, and put $F(y) = \sum_{k=1}^n b_k e^{\lambda_k y}$. Then F has finitely many real zeros, and $\operatorname{sp}(F(y_1), \ldots, F(y_j)) \preceq \operatorname{sp}(b)$ for all $y_1 < \cdots < y_j$. The same holds for ψF for every positive function ψ.*

Proof. Let $Z(F)$ be the number of distinct real zeros of F, and $S(b)$ the number of sign changes of b. We show $Z(F) \leq S(b)$ by induction on $S(b)$. If $S(b) = 0$, then F has the sign of the nonzero b_k everywhere. If $S(b) \geq 1$, choose i such that the last nonzero b_k with $k \leq i$ and the first nonzero b_k with $k > i$ have opposite signs, and choose $\lambda_i < \kappa < \lambda_{i+1}$. Then

$$G(y) = \frac{d}{dy}\big(e^{-\kappa y} F(y)\big) = \sum_{k=1}^n b_k(\lambda_k - \kappa) e^{(\lambda_k - \kappa)y}.$$

The coefficient $b_k(\lambda_k - \kappa)$ has the sign of $-b_k$ for $k \leq i$ and of b_k for $k > i$, so the coefficients of G have $S(b) - 1$ sign changes. By Rolle's theorem G vanishes between any two zeros of F, so $Z(F) \leq Z(G) + 1 \leq S(b)$.

If b_{k_0} is the first nonzero b_k, then $e^{-\lambda_{k_0} y} F(y) \to b_{k_0}$ as $y \to -\infty$. The zeros of F cut the line into at most $S(b) + 1$ open intervals, on each of which F has one sign. With equal neighbors merged, these signs form an alternating list that starts with the sign of b_{k_0} and has at most $S(b) + 1$ entries, so it is $\preceq \operatorname{sp}(b)$. The nonzero values $F(y_1), \ldots, F(y_j)$ lie in these intervals in order. A positive factor ψ changes no sign. □

Lemma F.2 (pointwise limits)**.** *Let v_n and v be real functions on $\mathbb{R}$ with $v_n(y) \to v(y)$ for every y, and let Π be an alternating list. If every v_n has values at most Π, then so has v. If every v_n has increments at most Π, then so has v.*

Proof. Let $y_1 < \cdots < y_j$, and let I be the set of indices i with $v(y_i) \neq 0$. For all large n, $v_n(y_i)$ has the sign of $v(y_i)$ for every $i \in I$. Then $\operatorname{sp}(v(y_1), \ldots, v(y_j))$ is the pattern of the list $(v_n(y_i))_{i \in I}$, which is a subsequence of $\operatorname{sp}(v_n(y_1), \ldots, v_n(y_j)) \preceq \Pi$. The increments of v_n at $y_1, \ldots, y_j$ tend to those of v, and the same argument applies to them. □

Lemma F.3 (one derivative fewer)**.** *Let u be twice continuously differentiable on $\mathbb{R}$, with $u'(y) \to 0$ as $y \to \pm\infty$. If u'' has values at most Π_5, then u' has values at most Π_4, and u has increments at most Π_4.*

Proof. It is enough to treat points $y_1 < \cdots < y_j$ at which the values of u' are nonzero and alternate in sign. Let s be the sign of $u'(y_1)$. Choose $y_0 < y_1$ with $|u'(y_0)| < |u'(y_1)|$, and $y_{j+1} > y_j$ with $|u'(y_{j+1})| < |u'(y_j)|$. For $0 \le i \le j$ the mean value theorem gives $w_i \in (y_i, y_{i+1})$ with $u''(w_i)(y_{i+1} - y_i) = u'(y_{i+1}) - u'(y_i)$. The right side has the sign s for $i = 0$, the sign of $u'(y_{i+1})$ for $1 \le i \le j-1$, and the sign of $-u'(y_j)$ for $i = j$. In each case this is $(-1)^i s$. So u'' takes $j+1$ alternating signs, starting with s, at $w_0 < \cdots < w_j$. This list is $\preceq \Pi_5$, so $j+1 \le 5$, and $j+1 \le 4$ if s is $-$. In both cases the list of the j signs of $u'(y_1), \dots, u'(y_j)$ is $\preceq \Pi_4$.

For the increments at $y_1 < \cdots < y_j$, each nonzero $u(y_{i+1}) - u(y_i)$ has the sign of $u'(\xi_i)$ for some $\xi_i \in (y_i, y_{i+1})$, by the mean value theorem. So the pattern of the increments is a pattern of values of u'. □

G The certificate for the threshold

We prove Theorem 4.4. This appendix states the steps, and the arXiv version of this paper [41] gives their full proofs. Fix $a \in (\frac{1}{2}, 1)$ and write $L = \mathcal{L}_a$ and $\phi(t) = \mathbb{E}e^{itL}$. Put $\theta_j = 1$ for even j and $\theta_j = a$ for odd j, let $m_0 = m_1 = 1$ and $m_n = (na - \theta_n)^{-1} \sum_{j=1}^{n-1} \theta_j m_j m_{n-j}$ for $n \ge 2$. Then $\mathbb{E}L^n = n!\, c_n$ with $c_n = m_n / \Gamma(1 + an)$ [8, Theorem 1.4]. As $\mathbb{E}e^{t|L|} < \infty$ for real t [8, Corollary 2.11], $\phi(t) = \sum_n c_n (it)^n$ for all real t. By [8, Theorem 1.1 and (2.10)], L has the law of $V^a L' + \xi(1-V)^a L''$, with V uniform on $[0,1]$, $\xi = \pm 1$, L' and L'' distributed as L, and all four independent. Since $|\phi(-s)| = |\phi(s)|$, this gives

$$|\phi(t)| \le \int_0^1 |\phi(tv^a)|\, |\phi(t(1-v)^a)|\, dv \qquad (t \in \mathbb{R}). \tag{8}$$

The density g_a of $\mathcal{L}_a^\circ$ has the characteristic function $\operatorname{Re}\phi$.

Step 1, one value of a. If $m_j \le AB^j$ for $1 \le j \le N$, with $A = ((N+1)a - 1)/N$, then $m_n \le AB^n$ for all $n \ge 1$, by induction, since $(na-1)/(n-1)$ increases in n. This bounds the tail of the moment series and of its derivatives by geometric series. Taylor's formula then bounds $|\phi|$ on cells covering $[0, T_3]$ by s_j, from finitely many coefficients. The next step turns these bounds into a bound for all t. Let $T_3 \ge 2^a T_2$, with T_2 an end point of a cell, and $\beta > 3$ with $a\beta > 1$. Let C be the largest $s_j r_j^\beta$ over the cells $[l_j, r_j]$ that meet $[T_2, T_3]$, and

$$R = T_2^{-1/a} \sum_{r_j \le T_2} s_j \Big(1 - \tfrac{1}{2}(r_j/T_2)^{1/a}\Big)^{-a\beta} \big(r_j^{1/a} - l_j^{1/a}\big) + \frac{2^{a\beta} C T_2^{-\beta}}{a\beta - 1}.$$

If $R < 1$, then $|\phi(t)| \le Ct^{-\beta}$ for all $t \ge T_3$ (the arXiv version of this paper [41]). The proof puts this bound into the right side of (8), gets back a smaller bound, and closes with a continuity argument. It is an explicit form of Step 4 in the proof of [8, Theorem 2.5]. By Fourier inversion g_a then has two continuous derivatives, and $g_a''(0) = -\frac{1}{\pi} \int_0^\infty t^2 \operatorname{Re}\phi(t)\, dt$. We integrate the moment series term by term on $[0, T_3]$, and the two error terms are explicit.

Step 2, intervals of a. With a as an interval, ball arithmetic fails, as the series cancels terms of size about 10^{60}. On $I = [a_0 - w, a_0 + w]$ we put $\rho = w/q$ with $0 < q < 1$ and expand in $a - a_0$. The coefficients are computed at the exact point a_0 by power series arithmetic. If h is analytic with $|h| \le M$ on the disc $|z - a_0| \le \rho$, then Cauchy's estimate bounds the remainder after degree K by $Mq^{K+1}/(1-q)$ on I. We bound M by enclosing $c_n(z)$ on a square that contains the disc, where nothing cancels, and beyond degree N by the complex form of Step 1 with the product formula for $|\Gamma(x)/\Gamma(x+iy)|^2$ [40, 5.8.3]. This gives a polynomial G and a number η with

$|g_a''(0) - G(a - a_0)| \le \eta$ for every $a \in I$. It also shows that $a \mapsto g_a''(0)$ is continuous on I (the arXiv version of this paper [41]).

Step 3, a near $\frac{1}{2}$. Put $y = 2a - 1$ and $\mu_n = m_n y^{\lfloor n/2 \rfloor}$. Then $\mu_2 = a$, and for $n \ge 3$ the recursion for μ_n has the factor y in its terms with n even and j odd, and no other change. So every μ_n is analytic for $\operatorname{Re} a > \frac{1}{4}$ (the arXiv version of this paper [41]). The variable $X = L/\sigma$ with $\sigma^2 = \mathbb{E}L^2 = 1/(y\Gamma(2a))$ has $\mathbb{E}X^n/n! = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor}/\Gamma(1 + an)$ for even n, and the same times $\sqrt{y\Gamma(2a)}$ for odd n. It satisfies the same fixed-point equation. Steps 1 and 2 in this variable reach $a = \frac{1}{2}$.

Step 4, the Hankel test. This step uses no Fourier analysis.

Lemma G.1 (a Hankel test)**.** *Let* $s_j = (2j+1)\mathbb{E}L^{2j}$. *Suppose that for some* $k \ge 1$ *and* $x \in \mathbb{R}^{k+1}$ *we have* $\sum_{i,j \le k} x_i x_j s_{i+j} < 0$ *or* $\sum_{i,j \le k} x_i x_j s_{i+j+1} < 0$. *Then case (a) of Theorem 4.2 fails, so case (b) holds.*

Proof. In case (a), $\mathcal{L}_a^\circ$ is unimodal about 0. By Khinchine's theorem [36] it has the law of UZ, with U uniform on $(0,1)$ and independent of Z. So $s_j = (2j+1)\mathbb{E}(\mathcal{L}_a^\circ)^{2j} = \mathbb{E}W^j$ with $W = Z^2 \ge 0$, and the two sums are $\mathbb{E}(\sum_i x_i W^i)^2 \ge 0$ and $\mathbb{E}W(\sum_i x_i W^i)^2 \ge 0$. □

The script `certify_threshold.py` checks every hypothesis in ball arithmetic [35]. It evaluates R with a as a ball over I and evaluates G on subintervals of I, bisecting until $G \pm \eta$ has one sign on each. For the Hankel test it takes k and a rational x from a floating-point eigenvector at the centre of a box of a, and checks the sum on the whole box with a Taylor model as in Step 2. Choices made in floating point, such as N, the precision and x, affect only whether a run succeeds. The cells have half-width $\frac{1}{32}$, the Taylor order in t is 12, $\beta = 4$, and the degree K is chosen so that $Mq^{K+1} \le e^{-60}$. Table 2 lists the runs.

range of a	method	pieces	parameters	time
$[\frac{1}{2}, 0.668]$	Steps 1 to 3	40	$q = \frac{1}{8}$, $T_2 \in [3, 10]$	2.1 h
$[0.668, 0.6712]$	Steps 1 and 2	8	$q = \frac{1}{32}$, $\rho \le \frac{1}{100}$, $T_2 \in [4, 10]$	12 min
$[0.6705, 1]$	Step 4	115	$k \le 56$	25 min

Table 2: The runs of the certificate. The times are summed over the processes of each run.

Proof of Theorem 4.4. The 40 intervals of the first run cover $[\frac{1}{2}, 0.668]$, and on each the enclosure of $g_a''(0)$ is negative. The 8 intervals of the second run cover $[0.668, 0.6712]$. On $[0.6688, 0.6692]$ we have $\eta < 1.1 \cdot 10^{-15}$. There the enclosure is negative on closed pieces that end at z_- and positive on closed pieces that start at z_+, and only (z_-, z_+) is undecided. Each other interval has one sign. So $g_a''(0) < 0$ on $(\frac{1}{2}, z_-]$ and $g_a''(0) > 0$ on $[z_+, 0.6712]$. The 115 boxes of the third run cover $[0.6705, 1]$, and on each the first sum of Lemma G.1 is negative.

By Step 1, g_a is even with two continuous derivatives, so $g_a'(0) = 0$. If $g_a''(0) < 0$, then $g_a' < 0$ on some $(0, \delta)$, which rules out case (b), where g_a is nondecreasing on $[0, c]$. If $g_a''(0) > 0$, then $g_a' > 0$ on some $(0, \delta)$, which rules out case (a). With Lemma G.1 on $[0.6705, 1)$ this proves parts (a) and (b). By Step 2, $a \mapsto g_a''(0)$ is continuous on $[0.6688, 0.6692]$, so it has a zero in (z_-, z_+). The signs above exclude zeros elsewhere in $(\frac{1}{2}, 0.6712]$. □

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